JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Matrices and Determinants

JEE Main Mathematics · Previous Year Questions with Answer Key

384	187	2002 – 2026	46	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

MCQ (Single Correct Answer) 267 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

If the system of linear equations :

$x + y + z = 6$
$x + 2y + 5z = 10$
$2x + 3y + \lambda z = \mu$

has infinitely many solutions, then the value of $\lambda + \mu$ equals:

(A) 12 (B) 16 (C) 22 (D) 28

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Let $A = \begin{bmatrix} \alpha & 1 & 2 \\ 2 & 3 & 0 \\ 0 & 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5\alpha & 0 \\ 0 & 4\alpha & -2\alpha \end{bmatrix} + \text{adj}(A)$. If $\det(B) = 66$, then $\det(\text{adj}(A))$ equals :

(A) 289 (B) 361 (C) 441 (D) 529

Q3. JEE Main 2026 (Online) 6th April Evening Shift

The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$x\cos 3\theta - 8y - 12z = 0$
$x\cos 2\theta + 3y + 3z = 0$
$x + y + 3z = 0$

has a non-trivial solution, is equal to :

(A) π (B) 2π (C) 3π (D) 4π

Q4. JEE Main 2026 (Online) 6th April Evening Shift

Let $\mathrm{A} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $\mathrm{B} = [\mathrm{b}_{ij}], 1 \le i, j \le 3$. If $\mathrm{B} = \mathrm{A}^{99} - \mathrm{I}$, then the value of $\frac{\mathrm{b}_{31} - \mathrm{b}_{21}}{\mathrm{b}_{32}}$ is :

(A) 99 (B) 199 (C) 149 (D) 159

Q5. JEE Main 2026 (Online) 5th April Evening Shift

If $f : \mathbf{N} \to \mathbf{Z}$ is defined by

$$f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}, k \in N,$$

and $\sum_{n=1}^{k} f(n) = 98$, then k is equal to :

(A) 3 (B) 4 (C) 5 (D) 6

Q6. JEE Main 2026 (Online) 5th April Evening Shift

Let M be a 3×3 matrix such that $\mathrm{M}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, \mathrm{M}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix}$ and $\mathrm{M}\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$. If $\mathrm{M}\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$, then $x+y+z$ equals :

(A) 4 (B) 5 (C) 7 (D) 11

Q7. JEE Main 2026 (Online) 5th April Morning Shift

Let A be a 3×3 matrix such that

$\mathrm{A}^{\mathrm{T}}\begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}5\\2\\2\end{bmatrix}, \mathrm{A}^{\mathrm{T}}\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}3\\1\\1\end{bmatrix}, \mathrm{A}\begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}3\\4\\4\end{bmatrix}$ and $\mathrm{A}\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\3\\1\end{bmatrix}$.

If $\det(A) = 1$, then $\det\left(\text{adj}\left(A^2 + A\right)\right)$ is equal to:

(A) 16 (B) 25 (C) 49 (D) 64

Q8. JEE Main 2026 (Online) 5th April Morning Shift

Consider the system of linear equations in x, y, z :

$x + 2y + tz = 0$
$6x + y + 5tz = 0$
$3x + t^2y + f(t)z = 0$

where $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function. If this system has infinitely many solutions for all $t \in \mathbb{R}$, then f

(A) is a constant function (B) is strictly increasing on $\mathbb{R}$
(C) is strictly decreasing on R (D) has two critical points

Q9. JEE Main 2026 (Online) 4th April Evening Shift

If the system of equations :

$x + y + z = 5$
$x + 2y + 3z = 9$
$x + 3y + \lambda z = \mu$

has infinitely many solutions, then the value of $\lambda + \mu$ is :

(A) 16 (B) 18 (C) 19 (D) 21

Q10. JEE Main 2026 (Online) 4th April Evening Shift

Let $A=\begin{bmatrix}1 & 2 & 7\\ 4 & -2 & 8\\ 3 & 8 & -7\end{bmatrix}$ and $\det(A-\alpha I)=0$, where α is a real number. If the largest possible value of α is p, then the circle $(x-p)^2+(y-2p)^2=320$, intersects the co-ordinate axes at

(A) 1 point (B) 2 points (C) 3 points (D) 4 points

Q11. JEE Main 2026 (Online) 4th April Morning Shift

Let $\mathrm{S}=\left\{\mathrm{A}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}: a,b,c,d\in\{0,1,2,3,4\} \text{ and } \mathrm{A}^2-4\,\mathrm{A}+3\mathrm{I}=0\right\}$ be a set of 2×2 matrices. Then the number of matrices in S , for which the sum of the diagonal elements is equal to 4 , is :

(A) 20 (B) 17 (C) 21 (D) 19

Q12. JEE Main 2026 (Online) 4th April Morning Shift

Let $A=\begin{bmatrix}1 & 1 & 2\\ -2 & 0 & 1\\ 1 & 3 & 5\end{bmatrix}$. Then the sum of all elements of the matrix $\operatorname{adj}\left(\operatorname{adj}\left(2(\operatorname{adj}\ \mathrm{A})^{-1}\right)\right)$ is equal to:

(A) 3 (B) 4 (C) -4 (D) -3

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

If the system of equations

$x+5y+6z=4$

$2x+3y+4z=7$

$x+6y+az=b$

has infinitely many solutions, then the point (a,b) lies on the line

(A) $y-x=3$ (B) $x-y=3$ (C) $x+y=11$ (D) $x+y=12$

Q14. JEE Main 2026 (Online) 2nd April Morning Shift

Let $\alpha,\beta\in\mathbb{R}$ be such that the system of linear equations

$$\begin{aligned}x+2y+z&=5\\ 2x+y+\alpha z&=5\\ 8x+4y+\beta z&=18\end{aligned}$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

(A) -4 (B) 4 (C) 8 (D) -8

Q15. JEE Main 2026 (Online) 2nd April Morning Shift

Let $A=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$. If $A^2-4A+I=O$ and $B^2-5B-6I=O$, then among the two statements:

(S1) : $[(B-A)(B+A)]^T=\begin{bmatrix}13 & 15\\ 7 & 10\end{bmatrix}$

and

(S2) : $\det(\text{adj}(A+B))=-5$

(A) only (S1) is correct

(B) only (S2) is correct

(C) both (S1) and (S2) are correct

(D) both (S1) and (S2) are wrong

Q16. JEE Main 2026 (Online) 28th January Morning Shift

Let A, B and C be three 2×2 matrices with real entries such that $B=(I+A)^{-1}$ and $\mathrm{A}+\mathrm{C}=\mathrm{I}$.

If $\mathrm{BC}=\begin{bmatrix}1 & -5\\ -1 & 2\end{bmatrix}$ and $\mathrm{CB}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}12\\ -6\end{bmatrix}$, then x_1+x_2 is

(A) 4

(B) 2

(C) 0

(D) -2

Q17. JEE Main 2026 (Online) 24th January Evening Shift

Let $P=[p_{ij}]$ and $Q=[q_{ij}]$ be two square matrices of order 3 such that $q_{\mathrm{ij}}=2^{(\mathrm{i}+\mathrm{j}-1)}\mathrm{p}_{\mathrm{ij}}$ and $\det(\mathrm{Q})=2^{10}$. Then the value of $\det(\text{adj}(\text{adj}\,\mathrm{P}))$ is:

(A) 81

(B) 16

(C) 124

(D) 32

Q18. JEE Main 2026 (Online) 24th January Evening Shift

Let $f(x)=\int\frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2}dx, x>0, \lim\limits_{x\to 0}f(x)=0$ and $f(1)=\frac{1}{4}$.

If $\mathrm{A}=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & f'(1) & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$ and $\mathrm{B}=\text{adj}(\text{adj}\,\mathrm{A})$ be such that $|\mathrm{B}|=81$, then α^2 is equal to

(A) 2

(B) 4

(C) 3

(D) 1

Q19. JEE Main 2026 (Online) 23rd January Evening Shift

The system of linear equations

$x + y + z = 6$

$2x + 5y + az = 36$

$x + 2y + 3z = b$

has :

(A) unique solution for $a = 8$ and $b = 16$

(B) infinitely many solutions for $a = 8$ and $b = 14$

(C) infinitely many solutions for $a = 8$ and $b = 16$

(D) unique solution for $a = 8$ and $b = 14$

Q20. JEE Main 2026 (Online) 23rd January Morning Shift

Among the statements :

I: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II: If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

(A) both are true (B) both are false (C) only I is true (D) only II is true

Q21. JEE Main 2026 (Online) 22nd January Evening Shift

Let n be the number obtained on rolling a fair die. If the probability that the system

$x - ny + z = 6$

$x + (n-2)y + (n+1)z = 8$

$(n-1)y + z = 1$

has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is :

(A) 22 (B) 20 (C) 24 (D) 21

Q22. JEE Main 2026 (Online) 22nd January Evening Shift

If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations $AX = B$, where $\text{adj } A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$, then $|x + y + z|$ is equal to :

(A) 3 (B) 2 (C) $\frac{3}{2}$ (D) 1

Q23. JEE Main 2026 (Online) 22nd January Morning Shift

If $\mathrm{A}=\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix}$, then the determinant of the matrix $\left(\mathrm{A}^{2025}-3\,\mathrm{A}^{2024}+\mathrm{A}^{2023}\right)$ is

(A) 12 (B) 24 (C) 28 (D) 16

Q24. JEE Main 2026 (Online) 21st January Evening Shift

If the system of equations

$3x+y+4z=3$

$2x+\alpha y-z=-3$

$x+2y+z=4$

has no solution, then the value of α is equal to:

(A) 13 (B) 4 (C) 19 (D) 23

Q25. JEE Main 2026 (Online) 21st January Evening Shift

For the matrices $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $B=\begin{bmatrix}-29 & 49\\ -13 & 18\end{bmatrix}$, if $(A^{15}+B)\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$, then among the following which one is **true**?

(A) $x=16, y=3$ (B) $x=5, y=7$ (C) $x=11, y=2$ (D) $x=18, y=11$

Q26. JEE Main 2025 (Online) 8th April Evening Shift

Let α be a solution of $x^2+x+1=0$, and for some *a* and *b* in $\mathbb{R}$,

$\begin{bmatrix}4 & a & b\end{bmatrix}\begin{bmatrix}1 & 16 & 13\\ -1 & -1 & 2\\ -2 & -14 & -8\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\end{bmatrix}$. If $\frac{4}{\alpha^4}+\frac{m}{\alpha^a}+\frac{n}{\alpha^b}=3$, then *m* + *n* is equal to _____

(A) 11 (B) 3 (C) 8 (D) 7

Q27. JEE Main 2025 (Online) 8th April Evening Shift

Let $A=\begin{bmatrix}2 & 2+p & 2+p+q\\ 4 & 6+2p & 8+3p+2q\\ 6 & 12+3p & 20+6p+3q\end{bmatrix}$.

If $\det(\text{adj}(\text{adj}(3A)))=2^m\cdot 3^n, m,n\in\mathbb{N}$, then $m+n$ is equal to

(A) 22 (B) 20 (C) 24 (D) 26

Q28. JEE Main 2025 (Online) 7th April Evening Shift

Let the system of equations

x + 5y - z = 1

4x + 3y - 3z = 7

24x + y + λz = μ

λ, μ ∈ $\mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system,

if x, y, z are integers and satisfy 7 ≤ x + y + z ≤ 77, is :

(A) 4 (B) 5 (C) 3 (D) 6

Q29. JEE Main 2025 (Online) 7th April Morning Shift

Let A be a 3×3 matrix such that $|\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A))| = 81$.

If $S = \left\{ n \in \mathbb{Z} : (|\operatorname{adj}(\operatorname{adj} A)|)^{\frac{(n-1)^2}{2}} = |A|^{(3n^2-5n-4)} \right\}$, then $\sum_{n \in S} \left| A^{(n^2+n)} \right|$ is equal to :

(A) 820 (B) 866 (C) 750 (D) 732

Q30. JEE Main 2025 (Online) 7th April Morning Shift

Let the system of equations :

$2x + 3y + 5z = 9$
$7x + 3y - 2z = 8$
$12x + 3y - (4 + \lambda)z = 16 - \mu$

have infinitely many solutions. Then the radius of the circle centred at (λ, μ) and touching the line $4x = 3y$ is :

(A) $\frac{7}{5}$ (B) $\frac{21}{5}$ (C) 7 (D) $\frac{17}{5}$

Q31. JEE Main 2025 (Online) 4th April Evening Shift

Let the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I$ for $n \geqslant 3$. Then the sum of all the elements of A^{50} is :

(A) 44 (B) 39 (C) 52 (D) 53

Q32. JEE Main 2025 (Online) 3rd April Morning Shift

Let A be a matrix of order 3×3 and $|A| = 5$. If $|2 \operatorname{adj}(3A \operatorname{adj}(2A))| = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma}, \alpha, \beta, \gamma \in N$, then $\alpha + \beta + \gamma$ is equal to

(A) 26 (B) 27 (C) 25 (D) 28

Q33. JEE Main 2025 (Online) 2nd April Evening Shift

If the system of equations

$2x + \lambda y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \mu z = 9$

has infinitely many solutions, then $(\lambda^2 + \mu^2)$ is equal to :

(A) 30 (B) 26 (C) 22 (D) 18

Q34. JEE Main 2025 (Online) 2nd April Evening Shift

Let A be a 3×3 real matrix such that $A^2(A - 2I) - 4(A - I) = O$, where I and O are the identity and null matrices, respectively. If $A^5 = \alpha A^2 + \beta A + \gamma I$, where α, β, and γ are real constants, then $\alpha + \beta + \gamma$ is equal to :

(A) 76 (B) 12 (C) 4 (D) 20

Q35. JEE Main 2025 (Online) 2nd April Morning Shift

Let $\mathrm{A} = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}, \alpha > 0$, such that $\det(\mathrm{A}) = 0$ and $\alpha + \beta = 1$. If I denotes 2×2 identity matrix, then the matrix $(I + A)^8$ is :

(A) $\begin{bmatrix} 257 & -64 \\ 514 & -127 \end{bmatrix}$ (B) $\begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$

(C) $\begin{bmatrix} 1025 & -511 \\ 2024 & -1024 \end{bmatrix}$ (D) $\begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$

Q36. JEE Main 2025 (Online) 2nd April Morning Shift

Let $a \in R$ and A be a matrix of order 3×3 such that $\det(A) = -4$ and $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$, where I is the identity matrix of order 3×3. If $\det((a + 1)\,\mathrm{adj}((a - 1)A))$ is $2^{\mathrm{m}}3^{\mathrm{n}}, \mathrm{m}, \mathrm{n} \in \{0, 1, 2, \ldots, 20\}$, then $\mathrm{m} + \mathrm{n}$ is equal to :

(A) 14 (B) 17 (C) 15 (D) 16

Q37. JEE Main 2025 (Online) 2nd April Morning Shift

If the system of linear equations

$3x + y + \beta z = 3$
$2x + \alpha y - z = -3$
$x + 2y + z = 4$

has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :

(A) 31 (B) 37 (C) 43 (D) 49

Q38. JEE Main 2025 (Online) 29th January Evening Shift

Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is:

(A) $\frac{5}{8}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) $\frac{3}{8}$

Q39. JEE Main 2025 (Online) 29th January Evening Shift

Let $\alpha, \beta\ (\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1$, $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}(n^{\alpha} + n^{\beta})$ is equal to :

(A) 3410 (B) 560 (C) 3080 (D) 440

Q40. JEE Main 2025 (Online) 29th January Evening Shift

Let $\mathrm{A} = [a_{ij}]$ be a matrix of order 3×3, with $a_{ij} = (\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}, \alpha, \beta \in \mathbf{Z}$, then $\alpha + \beta$ is equal to :

(A) 210 (B) 280 (C) 224 (D) 168

Q41. JEE Main 2025 (Online) 29th January Morning Shift

Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij}, $C_{ij} = \sum_{k=1}^{2} a_{ik}A_{jk}, 1 \leq i, j \leq 2$, and $C = [C_{ij}]$, then $8|C|$ is equal to :

(A) 288 (B) 262 (C) 222 (D) 242

Q42. JEE Main 2025 (Online) 29th January Morning Shift

Let M and m respectively be the maximum and the minimum values of

$$f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4\sin 4x \\ \sin^2 x & 1 + \cos^2 x & 4\sin 4x \\ \sin^2 x & \cos^2 x & 1 + 4\sin 4x \end{vmatrix}, x \in R$$

Then $M^4 - m^4$ is equal to :

(A) 1280 (B) 1040 (C) 1215 (D) 1295

Q43. JEE Main 2025 (Online) 28th January Evening Shift

Let $\mathrm{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{bmatrix}$ and $\mathrm{P} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \theta > 0$. If $\mathrm{B} = \mathrm{PAP}^{\top}, \mathrm{C} = \mathrm{P}^{\top}\mathrm{B}^{10}\mathrm{P}$ and the sum of the diagonal elements of C is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is :

(A) 127 (B) 2049 (C) 258 (D) 65

Q44. JEE Main 2025 (Online) 24th January Evening Shift

For some a, b, let $f(x) = \begin{vmatrix} \mathrm{a} + \frac{\sin x}{x} & 1 & \mathrm{b} \\ \mathrm{a} & 1 + \frac{\sin x}{x} & \mathrm{b} \\ \mathrm{a} & 1 & \mathrm{b} + \frac{\sin x}{x} \end{vmatrix}, x \neq 0, \lim_{x \to 0} f(x) = \lambda + \mu\mathrm{a} + \nu\mathrm{b}$. Then $(\lambda + \mu + v)^2$ is equal to :

(A) 25 (B) 16 (C) 9 (D) 36

Q45. JEE Main 2025 (Online) 24th January Evening Shift

If the system of equations

$x + 2y - 3z = 2$
$2x + \lambda y + 5z = 5$
$14x + 3y + \mu z = 33$

has infinitely many solutions, then $\lambda + \mu$ is equal to :

(A) 13 (B) 10 (C) 12 (D) 11

Q46. JEE Main 2025 (Online) 24th January Morning Shift

If the system of equations
$2x - y + z = 4$
$5x + \lambda y + 3z = 12$
$100x - 47y + \mu z = 212$
has infinitely many solutions, then $\mu - 2\lambda$ is equal to

(A) 56 (B) 59 (C) 57 (D) 55

Q47. JEE Main 2025 (Online) 23rd January Evening Shift

The system of equations
$x + y + z = 6,$
$x + 2y + 5z = 9,$
$x + 5y + \lambda z = \mu,$
has no solution if

(A) $\lambda = 17, \mu = 18$ (B) $\lambda = 17, \mu \neq 18$ (C) $\lambda = 15, \mu \neq 17$ (D) $\lambda \neq 17, \mu \neq 18$

Q48. JEE Main 2025 (Online) 23rd January Evening Shift

Let $A = [a_{ij}]$ be a 3×3 matrix such that $A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, A \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $A \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, then a_{23} equals :

(A) 2 (B) -1 (C) 1 (D) 0

Q49. JEE Main 2025 (Online) 23rd January Morning Shift

If the system of equations

$(\lambda-1)x+(\lambda-4)y+\lambda z=5$
$\lambda x+(\lambda-1)y+(\lambda-4)z=7$
$(\lambda+1)x+(\lambda+2)y-(\lambda+2)z=9$

has infinitely many solutions, then $\lambda^2+\lambda$ is equal to

(A) 20 (B) 10 (C) 6 (D) 12

Q50. JEE Main 2025 (Online) 23rd January Morning Shift

If $\mathrm{A}, \mathrm{B}, \text{and } \left(\text{adj } \left(\mathrm{A}^{-1}\right)+\text{adj } \left(\mathrm{B}^{-1}\right)\right)$ are non-singular matrices of same order, then the inverse of $A\left(\text{adj } \left(A^{-1}\right)+\text{adj } \left(B^{-1}\right)\right)^{-1} B$, is equal to

(A) $\frac{AB^{-1}}{|A|}+\frac{BA^{-1}}{|B|}$

(B) $\text{adj } \left(\mathrm{B}^{-1}\right)+\text{adj } \left(\mathrm{A}^{-1}\right)$

(C) $\mathrm{AB}^{-1}+\mathrm{A}^{-1}\,\mathrm{B}$

(D) $\frac{1}{|AB|}(\text{adj}(B)+\text{adj}(A))$

Q51. JEE Main 2025 (Online) 22nd January Evening Shift

If the system of linear equations :

$x+y+2z=6$
$2x+3y+\mathrm{az}=\mathrm{a}+1$
$-x-3y+\mathrm{b}z=2\,\mathrm{b}$

where $a, b \in \mathbf{R}$, has infinitely many solutions, then $7a+3b$ is equal to :

(A) 12 (B) 9 (C) 22 (D) 16

Q52. JEE Main 2025 (Online) 22nd January Evening Shift

For a 3×3 matrix M, let trace (M) denote the sum of all the diagonal elements of M. Let A be a 3×3 matrix such that $|A|=\frac{1}{2}$ and trace $(A)=3$. If $B=\text{adj}(\text{adj}(2A))$, then the value of $|B|+$ trace (B) equals :

(A) 56 (B) 132 (C) 174 (D) 280

Q53. JEE Main 2024 (Online) 9th April Evening Shift

Let $B=\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1}=A^{-1}$. If $BCB^{-1}=A$ and $C^4+\alpha C^2+\beta I=O$, then $2\beta-\alpha$ is equal to

(A) 16 (B) 10 (C) 8 (D) 2

Q54. JEE Main 2024 (Online) 9th April Morning Shift

Let $\lambda, \mu \in \mathbf{R}$. If the system of equations

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$

$97x + 155y - 189z = \mu$

has infinitely many solutions, then $\mu + 2\lambda$ is equal to :

(A) 24 (B) 25 (C) 27 (D) 22

Q55. JEE Main 2024 (Online) 8th April Evening Shift

If $\alpha \neq \mathrm{a}, \beta \neq \mathrm{b}, \gamma \neq \mathrm{c}$ and $\begin{vmatrix} \alpha & \mathrm{b} & \mathrm{c} \\ \mathrm{a} & \beta & \mathrm{c} \\ \mathrm{a} & \mathrm{b} & \gamma \end{vmatrix} = 0$, then $\frac{\mathrm{a}}{\alpha-\mathrm{a}} + \frac{\mathrm{b}}{\beta-\mathrm{b}} + \frac{\gamma}{\gamma-\mathrm{c}}$ is equal to :

(A) 2 (B) 3 (C) 1 (D) 0

Q56. JEE Main 2024 (Online) 8th April Evening Shift

If the system of equations $x + 4y - z = \lambda, 7x + 9y + \mu z = -3, 5x + y + 2z = -1$ has infinitely many solutions, then $(2\mu + 3\lambda)$ is equal to :

(A) -2 (B) 2 (C) 3 (D) -3

Q57. JEE Main 2024 (Online) 8th April Morning Shift

Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3, then $2a + 3b$ is equal to

(A) -10 (B) -12 (C) -13 (D) -9

Q58. JEE Main 2024 (Online) 6th April Evening Shift

If A is a square matrix of order 3 such that $\det(A) = 3$ and $\det\left(\operatorname{adj}\left(-4\operatorname{adj}\left(-3\operatorname{adj}\left(3\operatorname{adj}\left((2\,\mathrm{A})^{-1}\right)\right)\right)\right)\right) = 2^{\mathrm{m}}3^{\mathrm{n}}$, then $\mathrm{m} + 2\mathrm{n}$ is equal to :

(A) 2 (B) 4 (C) 3 (D) 6

Q59. JEE Main 2024 (Online) 6th April Morning Shift

For $\alpha, \beta \in \mathbb{R}$ and a natural number n, let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10} - A_8$ is

(A) $4\alpha + 2\beta$ (B) 0 (C) $2n$ (D) $2\alpha + 4\beta$

Q60. JEE Main 2024 (Online) 5th April Evening Shift

The values of m, n, for which the system of equations

$x + y + z = 4,$
$2x + 5y + 5z = 17,$
$x + 2y + \mathrm{m}z = \mathrm{n}$

has infinitely many solutions, satisfy the equation :

(A) $\mathrm{m}^2 + \mathrm{n}^2 - \mathrm{m} - \mathrm{n} = 46$ (B) $\mathrm{m}^2 + \mathrm{n}^2 + \mathrm{mn} = 68$

(C) $\mathrm{m}^2 + \mathrm{n}^2 - \mathrm{mn} = 39$ (D) $\mathrm{m}^2 + \mathrm{n}^2 + \mathrm{m} + \mathrm{n} = 64$

Q61. JEE Main 2024 (Online) 5th April Evening Shift

Let $\alpha\beta \neq 0$ and $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$. If $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ is the matrix of cofactors of the elements of A, then $\det(AB)$ is equal to :

(A) 64 (B) 343 (C) 125 (D) 216

Q62. JEE Main 2024 (Online) 5th April Morning Shift

Let A and B be two square matrices of order 3 such that $|\mathrm{A}| = 3$ and $|\mathrm{B}| = 2$. Then $|\mathrm{A}^{\mathrm{T}}\mathrm{A}(\mathrm{adj}(2\,\mathrm{A}))^{-1}(\mathrm{adj}(4\,\mathrm{B}))(\mathrm{adj}(\mathrm{AB}))^{-1}\mathrm{AA}^{\mathrm{T}}|$ is equal to :

(A) 32 (B) 81 (C) 64 (D) 108

Q63. JEE Main 2024 (Online) 5th April Morning Shift

If the system of equations

$11x + y + \lambda z = -5$
$2x + 3y + 5z = 3$
$8x - 19y - 39z = \mu$

has infinitely many solutions, then $\lambda^4 - \mu$ is equal to :

(A) 51 (B) 45 (C) 47 (D) 49

Q64. JEE Main 2024 (Online) 4th April Evening Shift

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = I + \mathrm{adj}(A) + (\mathrm{adj}\,A)^2 + \ldots + (\mathrm{adj}\,A)^{10}$. Then, the sum of all the elements of the matrix B is:

(A) -110 (B) 22 (C) -124 (D) -88

Q65. JEE Main 2024 (Online) 4th April Morning Shift

Let $\alpha \in (0, \infty)$ and $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. If $\det\left(\text{adj}\left(2A - A^T\right) \cdot \text{adj}\left(A - 2A^T\right)\right) = 2^8$, then $(\det(A))^2$ is equal to:

(A) 16 (B) 36 (C) 49 (D) 1

Q66. JEE Main 2024 (Online) 4th April Morning Shift

If the system of equations

$x + (\sqrt{2}\sin\alpha)y + (\sqrt{2}\cos\alpha)z = 0$
$x + (\cos\alpha)y + (\sin\alpha)z = 0$
$x + (\sin\alpha)y - (\cos\alpha)z = 0$

has a non-trivial solution, then $\alpha \in \left(0, \frac{\pi}{2}\right)$ is equal to :

(A) $\frac{5\pi}{24}$ (B) $\frac{11\pi}{24}$ (C) $\frac{7\pi}{24}$ (D) $\frac{3\pi}{4}$

Q67. JEE Main 2024 (Online) 1st February Evening Shift

Let the system of equations $x + 2y + 3z = 5, 2x + 3y + z = 9, 4x + 3y + \lambda z = \mu$ have infinite number of solutions. Then $\lambda + 2\mu$ is equal to :

(A) 22 (B) 17 (C) 15 (D) 28

Q68. JEE Main 2024 (Online) 1st February Morning Shift

If $\mathrm{A} = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}, \mathrm{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \mathrm{C} = \mathrm{ABA}^{\mathrm{T}}$ and $\mathrm{X} = \mathrm{A}^{\mathrm{T}}\mathrm{C}^2\,\mathrm{A}$, then $\det \mathrm{X}$ is equal to :

(A) 243 (B) 729 (C) 27 (D) 891

Q69. JEE Main 2024 (Online) 1st February Morning Shift

If the system of equations

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

has infinitely many solutions, then $13\alpha\beta$ is equal to :

(A) 1110 (B) 1120 (C) 1210 (D) 1220

Q70. JEE Main 2024 (Online) 31st January Evening Shift

Let A be a 3×3 real matrix such that

$$A\begin{pmatrix}1\\0\\1\end{pmatrix} = 2\begin{pmatrix}1\\0\\1\end{pmatrix}, A\begin{pmatrix}-1\\0\\1\end{pmatrix} = 4\begin{pmatrix}-1\\0\\1\end{pmatrix}, A\begin{pmatrix}0\\1\\0\end{pmatrix} = 2\begin{pmatrix}0\\1\\0\end{pmatrix}.$$

Then, the system $(A - 3I)\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}$ has :

(A) exactly two solutions

(B) infinitely many solutions

(C) unique solution

(D) no solution

Q71. JEE Main 2024 (Online) 31st January Morning Shift

If the system of linear equations

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

has infinitely many solutions, then $12\alpha + 13\beta$ is equal to

(A) 60 (B) 54 (C) 64 (D) 58

Q72. JEE Main 2024 (Online) 30th January Evening Shift

Let $R = \begin{pmatrix}x & 0 & 0\\0 & y & 0\\0 & 0 & z\end{pmatrix}$ be a non-zero 3×3 matrix, where

$x\sin\theta = y\sin\left(\theta + \frac{2\pi}{3}\right) = z\sin\left(\theta + \frac{4\pi}{3}\right) \neq 0, \theta \in (0, 2\pi)$. For a square matrix M, let trace (M) denote the sum of all the diagonal entries of M. Then, among the statements:

(I) Trace $(R) = 0$

(II) If trace $(\text{adj}(\text{adj}(R))) = 0$, then R has exactly one non-zero entry.

(A) Only (I) is true

(B) Only (II) is true

(C) Both (I) and (II) are true

(D) Neither (I) nor (II) is true

Q73. JEE Main 2024 (Online) 30th January Evening Shift

Consider the system of linear equations $x + y + z = 5, x + 2y + \lambda^2 z = 9, x + 3y + \lambda z = \mu$, where $\lambda, \mu \in \mathbb{R}$. Then, which of the following statements is NOT correct?

(A) System is consistent if $\lambda \neq 1$ and $\mu = 13$

(B) System is inconsistent if $\lambda = 1$ and $\mu \neq 13$

(C) System has unique solution if $\lambda \neq 1$ and $\mu \neq 13$

(D) System has infinite number of solutions if $\lambda = 1$ and $\mu = 13$

Q74. JEE Main 2024 (Online) 30th January Morning Shift

Consider the system of linear equations
$x+y+z=4\mu, x+2y+2\lambda z=10\mu, x+3y+4\lambda^2 z=\mu^2+15$ where $\lambda, \mu \in \mathbf{R}$. Which one of the following statements is NOT correct ?

(A) The system has unique solution if $\lambda \neq \frac{1}{2}$ and $\mu \neq 1, 15$

(B) The system has infinite number of solutions if $\lambda = \frac{1}{2}$ and $\mu = 15$

(C) The system is consistent if $\lambda \neq \frac{1}{2}$

(D) The system is inconsistent if $\lambda = \frac{1}{2}$ and $\mu \neq 1$

Q75. JEE Main 2024 (Online) 29th January Evening Shift

Let $A=\begin{bmatrix}2 & 1 & 2\\ 6 & 2 & 11\\ 3 & 3 & 2\end{bmatrix}$ and $P=\begin{bmatrix}1 & 2 & 0\\ 5 & 0 & 2\\ 7 & 1 & 5\end{bmatrix}$. The sum of the prime factors of $\left|P^{-1}AP-2I\right|$ is equal to

(A) 66 (B) 27 (C) 23 (D) 26

Q76. JEE Main 2024 (Online) 29th January Morning Shift

Let $A=\begin{bmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{bmatrix}$ and $|2\,\mathrm{A}|^3=2^{21}$ where $\alpha, \beta \in Z$. Then a value of α is

(A) 9 (B) 17 (C) 3 (D) 5

Q77. JEE Main 2024 (Online) 29th January Morning Shift

Let A be a square matrix such that $\mathrm{AA}^{\mathrm{T}}=\mathrm{I}$. Then $\frac{1}{2}A\left[\left(A+A^T\right)^2+\left(A-A^T\right)^2\right]$ is equal to

(A) $\mathrm{A}^2+\mathrm{A}^{\mathrm{T}}$ (B) $\mathrm{A}^3+\mathrm{I}$

(C) $\mathrm{A}^3+\mathrm{A}^{\mathrm{T}}$ (D) $\mathrm{A}^2+\mathrm{I}$

Q78. JEE Main 2024 (Online) 27th January Evening Shift

The values of α, for which $\begin{vmatrix}1 & \frac{3}{2} & \alpha+\frac{3}{2}\\ 1 & \frac{1}{3} & \alpha+\frac{1}{3}\\ 2\alpha+3 & 3\alpha+1 & 0\end{vmatrix}=0$, lie in the interval

(A) $(-2,1)$ (B) $\left(-\frac{3}{2},\frac{3}{2}\right)$

(C) $(-3,0)$ (D) $(0,3)$

Q79. JEE Main 2024 (Online) 27th January Morning Shift

Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Given below are two statements :

Statement I : $f(-x)$ is the inverse of the matrix $f(x)$.

Statement II : $f(x)f(y) = f(x+y)$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q80. JEE Main 2023 (Online) 13th April Evening Shift

If the system of equations

$2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

has infinitely many solutions, then $(\lambda + \mu)^2 + (\lambda - \mu)^2$ is equal to

(A) 916 (B) 912 (C) 920 (D) 904

Q81. JEE Main 2023 (Online) 13th April Morning Shift

For the system of linear equations

$2x + 4y + 2az = b$

$x + 2y + 3z = 4$

$2x - 5y + 2z = 8$

which of the following is NOT correct?

(A) It has infinitely many solutions if $a = 3, b = 8$

(B) It has infinitely many solutions if $a = 3, b = 6$

(C) It has unique solution if $a = b = 8$

(D) It has unique solution if $a = b = 6$

Q82. JEE Main 2023 (Online) 13th April Morning Shift

Let $B=\begin{bmatrix}1 & 3 & \alpha\\ 1 & 2 & 3\\ \alpha & \alpha & 4\end{bmatrix}, \alpha>2$ be the adjoint of a matrix A and $|A|=2$. Then $\begin{bmatrix}\alpha & -2\alpha & \alpha\end{bmatrix}B\begin{bmatrix}\alpha\\ -2\alpha\\ \alpha\end{bmatrix}$ is equal to :

(A) 32 (B) -16 (C) 0 (D) 16

Q83. JEE Main 2023 (Online) 13th April Morning Shift

The number of symmetric matrices of order 3, with all the entries from the set $\{0,1,2,3,4,5,6,7,8,9\}$ is :

(A) 10^9 (B) 9^{10} (C) 10^6 (D) 6^{10}

Q84. JEE Main 2023 (Online) 12th April Morning Shift

Let $A=\begin{bmatrix}1 & \frac{1}{51}\\ 0 & 1\end{bmatrix}$. If $B=\begin{bmatrix}1 & 2\\ -1 & -1\end{bmatrix}A\begin{bmatrix}-1 & -2\\ 1 & 1\end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50}B^n$ is equal to

(A) 50 (B) 75 (C) 100 (D) 125

Q85. JEE Main 2023 (Online) 11th April Evening Shift

If the system of linear equations

$7x+11y+\alpha z=13$

$5x+4y+7z=\beta$

$175x+194y+57z=361$

has infinitely many solutions, then $\alpha+\beta+2$ is equal to :

(A) 6 (B) 4 (C) 5 (D) 3

Q86. JEE Main 2023 (Online) 11th April Evening Shift

$\begin{vmatrix}x+1 & x & x\\ x & x+\lambda & x\\ x & x & x+\lambda^2\end{vmatrix}=\frac{9}{8}(103x+81)$, then λ, $\frac{\lambda}{3}$ are the roots of the equation :

(A) $4x^2+24x-27=0$ (B) $4x^2-24x+27=0$

(C) $4x^2-24x-27=0$ (D) $4x^2+24x+27=0$

Q87. JEE Main 2023 (Online) 11th April Morning Shift

Let A be a 2×2 matrix with real entries such that $\mathrm{A}'=\alpha\mathrm{A}+\mathrm{I}$, where $\alpha\in\mathbb{R}-\{-1,1\}$. If $\det\left(A^2-A\right)=4$, then the sum of all possible values of α is equal to :

(A) 2 (B) $\frac{3}{2}$ (C) 0 (D) $\frac{5}{2}$

Q88. JEE Main 2023 (Online) 10th April Morning Shift

For the system of linear equations

$2x - y + 3z = 5$

$3x + 2y - z = 7$

$4x + 5y + \alpha z = \beta,$

which of the following is **NOT** correct?

(A) The system has infinitely many solutions for $\alpha = -6$ and $\beta = 9$

(B) The system has a unique solution for $\alpha \neq -5$ and $\beta = 8$

(C) The system is inconsistent for $\alpha = -5$ and $\beta = 8$

(D) The system has infinitely many solutions for $\alpha = -5$ and $\beta = 9$

Q89. JEE Main 2023 (Online) 8th April Evening Shift

If $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $\mathrm{A}^{-1} = \alpha \mathrm{A} + \beta \mathrm{I}$ and $\alpha + \beta = -2$, then $4\alpha^2 + \beta^2 + \lambda^2$ is equal to :

(A) 12 (B) 10 (C) 19 (D) 14

Q90. JEE Main 2023 (Online) 8th April Evening Shift

Let S be the set of all values of $\theta \in [-\pi, \pi]$ for which the system of linear equations

$x + y + \sqrt{3}z = 0$

$-x + (\tan\theta)y + \sqrt{7}z = 0$

$x + y + (\tan\theta)z = 0$

has non-trivial solution. Then $\frac{120}{\pi} \sum_{\theta \in \mathrm{S}} \theta$ is equal to :

(A) 40 (B) 30 (C) 10 (D) 20

Q91. JEE Main 2023 (Online) 8th April Morning Shift

Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. If $|\mathrm{adj}(\mathrm{adj}(\mathrm{adj}\, 2A))| = (16)^n$, then n is equal to :

(A) 9 (B) 8 (C) 10 (D) 12

Q92. JEE Main 2023 (Online) 8th April Morning Shift

Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$. If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b - 3c - 4d$ equal to :

(A) 2004 (B) 2006 (C) 2007 (D) 2005

Q93. JEE Main 2023 (Online) 6th April Evening Shift

Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to :

(A) 18 (B) 22 (C) 24 (D) 40

Q94. JEE Main 2023 (Online) 6th April Evening Shift

For the system of equations

$x + y + z = 6$

$x + 2y + \alpha z = 10$

$x + 3y + 5z = \beta$, which one of the following is **NOT** true?

(A) System has a unique solution for $\alpha = 3, \beta \neq 14$.

(B) System has infinitely many solutions for $\alpha = 3, \beta = 14$.

(C) System has no solution for $\alpha = 3, \beta = 24$.

(D) System has a unique solution for $\alpha = -3, \beta = 14$.

Q95. JEE Main 2023 (Online) 6th April Morning Shift

If the system of equations

$x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

has infinitely many solutions, then $2a + 3b$ is equal to :

(A) 28 (B) 25 (C) 20 (D) 23

Q96. JEE Main 2023 (Online) 6th April Morning Shift

Let $\mathrm{A} = [\mathrm{a_{ij}}]_{2\times 2}$, where $\mathrm{a_{ij}} \neq 0$ for all i, j and $\mathrm{A}^2 = \mathrm{I}$. Let a be the sum of all diagonal elements of A and $\mathrm{b} = |\mathrm{A}|$. Then $3a^2 + 4b^2$ is equal to :

(A) 4 (B) 3 (C) 14 (D) 7

Q97. JEE Main 2023 (Online) 1st February Evening Shift

For the system of linear equations $\alpha x + y + z = 1, x + \alpha y + z = 1, x + y + \alpha z = \beta$, which one of the following statements is **NOT** correct?

(A) It has infinitely many solutions if $\alpha = 1$ and $\beta = 1$

(B) It has infinitely many solutions if $\alpha = 2$ and $\beta = -1$

(C) $x + y + z = \frac{3}{4}$ if $\alpha = 2$ and $\beta = 1$

(D) It has no solution if $\alpha = -2$ and $\beta = 1$

Q98. JEE Main 2023 (Online) 1st February Evening Shift

If $A=\frac{1}{2}\begin{bmatrix}1 & \sqrt{3}\\ -\sqrt{3} & 1\end{bmatrix}$, then :

(A) $A^{30}-A^{25}=2I$ (B) $A^{30}+A^{25}-A=I$ (C) $A^{30}=A^{25}$ (D) $A^{30}+A^{25}+A=I$

Q99. JEE Main 2023 (Online) 1st February Morning Shift

Let S denote the set of all real values of λ such that the system of equations

$\lambda x+y+z=1$

$x+\lambda y+z=1$

$x+y+\lambda z=1$

is inconsistent, then $\sum_{\lambda\in S}\left(|\lambda|^2+|\lambda|\right)$ is equal to

(A) 12 (B) 2 (C) 4 (D) 6

Q100. JEE Main 2023 (Online) 1st February Morning Shift

Let $f(x)=\begin{vmatrix}1+\sin^2x & \cos^2x & \sin 2x\\ \sin^2x & 1+\cos^2x & \sin 2x\\ \sin^2x & \cos^2x & 1+\sin 2x\end{vmatrix}$, $x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f, then

(A) $\alpha^2-\beta^2=4\sqrt{3}$ (B) $\beta^2-2\sqrt{\alpha}=\frac{19}{4}$

(C) $\beta^2+2\sqrt{\alpha}=\frac{19}{4}$ (D) $\alpha^2+\beta^2=\frac{9}{2}$

Q101. JEE Main 2023 (Online) 31st January Morning Shift

For the system of linear equations

$x+y+z=6$

$\alpha x+\beta y+7z=3$

$x+2y+3z=14$

which of the following is **NOT** true ?

(A) If $\alpha=\beta=7$, then the system has no solution

(B) For every point $(\alpha,\beta)\neq(7,7)$ on the line $x-2y+7=0$, the system has infinitely many solutions

(C) There is a unique point (α,β) on the line $x+2y+18=0$ for which the system has infinitely many solutions

(D) If $\alpha=\beta$ and $\alpha\neq 7$, then the system has a unique solution

Q102. JEE Main 2023 (Online) 31st January Morning Shift

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to :

(A) 4094 (B) 2050 (C) 6144 (D) 4097

Q103. JEE Main 2023 (Online) 30th January Evening Shift

For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$x - y + z = 5$
$2x + 2y + \alpha z = 8$
$3x - y + 4z = \beta$

has infinitely many solutions. Then α and β are the roots of :

(A) $x^2 + 18x + 56 = 0$ (B) $x^2 - 10x + 16 = 0$ (C) $x^2 + 14x + 24 = 0$ (D) $x^2 - 18x + 56 = 0$

Q104. JEE Main 2023 (Online) 30th January Morning Shift

Let the system of linear equations

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

have infinitely many solutions. Then the system

$(k + 1)x + (2k - 1)y = 7$

$(2k + 1)x + (k + 5)y = 10$

has :

(A) unique solution satisfying $x - y = 1$ (B) infinitely many solutions

(C) no solution (D) unique solution satisfying $x + y = 1$

Q105. JEE Main 2023 (Online) 29th January Morning Shift

Let α and β be real numbers. Consider a 3 $\times$ 3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then

(A) $\alpha = 1$ (B) $\alpha = 4$ (C) $\beta = 8$ (D) $\beta = -8$

Q106. JEE Main 2023 (Online) 29th January Morning Shift

Consider the following system of equations

$\alpha x + 2y + z = 1$

$2\alpha x + 3y + z = 1$

$3x + \alpha y + 2z = \beta$

for some $\alpha, \beta \in \mathbb{R}$. Then which of the following is NOT correct.

(A) It has a solution for all $\alpha \neq -1$ and $\beta = 2$

(B) It has no solution if $\alpha = -1$ and $\beta \neq 2$

(C) It has no solution for $\alpha = -1$ and for all $\beta \in \mathbb{R}$

(D) It has no solution for $\alpha = 3$ and for all $\beta \neq 2$

Q107. JEE Main 2023 (Online) 25th January Evening Shift

Let A, B, C be 3 $\times$ 3 matrices such that A is symmetric and B and C are skew-symmetric. Consider the statements

(S1) $A^{13} B^{26} - B^{26} A^{13}$ is symmetric

(S2) $A^{26} C^{13} - C^{13} A^{26}$ is symmetric

Then,

(A) Only S2 is true

(B) Only S1 is true

(C) Both S1 and S2 are false

(D) Both S1 and S2 are true

Q108. JEE Main 2023 (Online) 25th January Evening Shift

Let $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where $i = \sqrt{-1}$. If $M = A^T BA$, then the inverse of the matrix $AM^{2023}A^T$ is

(A) $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

Q109. JEE Main 2023 (Online) 25th January Morning Shift

Let S_1 and S_2 be respectively the sets of all $a \in \mathbb{R} - \{0\}$ for which the system of linear equations

$ax + 2ay - 3az = 1$

$(2a+1)x + (2a+3)y + (a+1)z = 2$

$(3a+5)x + (a+5)y + (a+2)z = 3$

has unique solution and infinitely many solutions. Then

(A) $n(S_1) = 2$ and S_2 is an infinite set

(B) $S_1 = \Phi$ and $S_2 = \mathbb{R} - \{0\}$

(C) $S_1 = \mathbb{R} - \{0\}$ and $S_2 = \Phi$

(D) S_1 is an infinite set and $n(S_2) = 2$

Q110. JEE Main 2023 (Online) 24th January Evening Shift

If the system of equations

$x + 2y + 3z = 3$

$4x + 3y - 4z = 4$

$8x + 4y - \lambda z = 9 + \mu$

has infinitely many solutions, then the ordered pair (λ, μ) is equal to :

(A) $\left(\frac{72}{5}, \frac{21}{5}\right)$ (B) $\left(-\frac{72}{5}, -\frac{21}{5}\right)$

(C) $\left(-\frac{72}{5}, \frac{21}{5}\right)$ (D) $\left(\frac{72}{5}, -\frac{21}{5}\right)$

Q111. JEE Main 2023 (Online) 24th January Morning Shift

If A and B are two non-zero n $\times$ n matrices such that $\mathrm{A}^2 + \mathrm{B} = \mathrm{A}^2\mathrm{B}$, then :

(A) $\mathrm{A}^2\mathrm{B} = \mathrm{I}$ (B) $\mathrm{A}^2 = \mathrm{I}$ or $\mathrm{B} = \mathrm{I}$

(C) $\mathrm{A}^2\mathrm{B} = \mathrm{B}\mathrm{A}^2$ (D) $\mathrm{AB} = \mathrm{I}$

Q112. JEE Main 2022 (Online) 29th July Evening Shift

Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation ?

(A) $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Q113. JEE Main 2022 (Online) 29th July Evening Shift

If the system of equations

$x + y + z = 6$
$2x + 5y + \alpha z = \beta$
$x + 2y + 3z = 14$

has infinitely many solutions, then $\alpha + \beta$ is equal to

(A) 8 (B) 36 (C) 44 (D) 48

Q114. JEE Main 2022 (Online) 29th July Morning Shift

Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

(A) the system of linear equations $AX = 0$ has a unique solution

(B) the system of linear equations $AX = 0$ has infinitely many solutions

(C) B is an invertible matrix

(D) $\mathrm{adj}(\mathrm{A})$ is an invertible matrix

Q115. JEE Main 2022 (Online) 28th July Evening Shift

Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?

(A) $\mathrm{A}^4 - \mathrm{B}^4$ is a symmetric matrix

(B) $\mathrm{AB} - \mathrm{BA}$ is a symmetric matrix

(C) $\mathrm{B}^5 - \mathrm{A}^5$ is a skew-symmetric matrix

(D) $\mathrm{AB} + \mathrm{BA}$ is a skew-symmetric matrix

Q116. JEE Main 2022 (Online) 27th July Morning Shift

Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to

(A) -10 (B) -6 (C) 6 (D) 10

Q117. JEE Main 2022 (Online) 26th July Evening Shift

Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of $A'BA$ is:

(A) 1224 (B) 1042 (C) 540 (D) 539

Q118. JEE Main 2022 (Online) 26th July Morning Shift

If the system of linear equations.

$8x + y + 4z = -2$

$x + y + z = 0$

$\lambda x - 3y = \mu$

has infinitely many solutions, then the distance of the point $\left(\lambda, \mu, -\frac{1}{2}\right)$ from the plane $8x + y + 4z + 2 = 0$ is :

(A) $3\sqrt{5}$ (B) 4 (C) $\frac{26}{9}$ (D) $\frac{10}{3}$

Q119. JEE Main 2022 (Online) 25th July Evening Shift

The number of real values of λ, such that the system of linear equations

$2x - 3y + 5z = 9$

$x + 3y - z = -18$

$3x - y + (\lambda^2 - |\lambda|)z = 16$

has no solutions, is

(A) 0 (B) 1 (C) 2 (D) 4

Q120. JEE Main 2022 (Online) 25th July Morning Shift

The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$3(\sin 3\theta)x - y + z = 2$

$3(\cos 2\theta)x + 4y + 3z = 3$

$6x + 7y + 7z = 9$

has no solution, is :

(A) 6 (B) 7 (C) 8 (D) 9

Q121. JEE Main 2022 (Online) 30th June Morning Shift

Let $A = \begin{bmatrix} 1 & -2 & \alpha \\ \alpha & 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & \alpha \\ -1 & 2 \\ 4 & -5 \end{bmatrix}$, $\alpha \in C$. Then the absolute value of the sum of all values of α for which det(AB) = 0 is :

(A) 3 (B) 4 (C) 2 (D) 5

Q122. JEE Main 2022 (Online) 29th June Morning Shift

If the system of linear equations

2x + y − z = 7

x − 3y + 2z = 1

x + 4y + δz = k, where δ, k ∈ R has infinitely many solutions, then δ + k is equal to:

(A) −3 (B) 3 (C) 6 (D) 9

Q123. JEE Main 2022 (Online) 29th June Morning Shift

Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all i, j = 1, 2, 3. Then, the matrix $A^2 + A^3 + \ldots\ldots + A^{10}$ is equal to :

(A) $\left(\frac{3^{10}-3}{2}\right)A$ (B) $\left(\frac{3^{10}-1}{2}\right)A$

(C) $\left(\frac{3^{10}+1}{2}\right)A$ (D) $\left(\frac{3^{10}+3}{2}\right)A$

Q124. JEE Main 2022 (Online) 28th June Morning Shift

If the system of linear equations

$2x + 3y - z = -2$

$x + y + z = 4$

$x - y + |\lambda|z = 4\lambda - 4$

where, $\lambda \in$ R, has no solution, then

(A) λ = 7 (B) $\lambda = -7$ (C) λ = 8 (D) λ^2 = 1

Q125. JEE Main 2022 (Online) 27th June Morning Shift

Let the system of linear equations
$x + 2y + z = 2$,
$\alpha x + 3y - z = \alpha$,
$-\alpha x + y + 2z = -\alpha$
be inconsistent. Then α is equal to :

(A) $\frac{5}{2}$ (B) $-\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $-\frac{7}{2}$

Q126. JEE Main 2022 (Online) 26th June Evening Shift

If the system of equations

αx + y + z = 5, x + 2y + 3z = 4, x + 3y + 5z = β

has infinitely many solutions, then the ordered pair (α, β) is equal to :

(A) (1, -3) (B) $(-1, 3)$ (C) (1, 3) (D) $(-1, -3)$

Q127. JEE Main 2022 (Online) 26th June Morning Shift

The ordered pair (a, b), for which the system of linear equations

3x $-$ 2y + z = b

5x $-$ 8y + 9z = 3

2x + y + az = -1

has no solution, is :

(A) $\left(3, \frac{1}{3}\right)$ (B) $\left(-3, \frac{1}{3}\right)$
(C) $\left(-3, -\frac{1}{3}\right)$ (D) $\left(3, -\frac{1}{3}\right)$

Q128. JEE Main 2022 (Online) 25th June Evening Shift

The system of equations

$-kx + 3y - 14z = 25$

$-15x + 4y - kz = 3$

$-4x + y + 3z = 4$

is consistent for all k in the set

(A) R (B) R $-$ {-11, 13} (C) R $-$ {13} (D) R $-$ {-11, 11}

Q129. JEE Main 2022 (Online) 25th June Morning Shift

Let A be a 3×3 real matrix such that

$A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}; A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix}$ and $A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}$.

If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A - 2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$ has :

(A) no solution

(B) infinitely many solutions

(C) unique solution

(D) exactly two solutions

Q130. JEE Main 2022 (Online) 25th June Morning Shift

Let $A = \begin{bmatrix}0 & -2\\2 & 0\end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is :

(A) a non-identity symmetric matrix

(B) a skew-symmetric matrix

(C) neither symmetric nor skew-symmetric matrix

(D) an identity matrix

Q131. JEE Main 2022 (Online) 24th June Evening Shift

Let the system of linear equations

x + y + αz = 2

3x + y + z = 4

x + 2z = 1

have a unique solution (x^*, y^*, z^*). If (α, x^*), (y^*, α) and (x^*, $-y^*$) are collinear points, then the sum of absolute values of all possible values of α is

(A) 4 (B) 3 (C) 2 (D) 1

Q132. JEE Main 2022 (Online) 24th June Morning Shift

The number of values of α for which the system of equations :

x + y + z = α

αx + 2αy + 3z = -1

x + 3αy + 5z = 4

is inconsistent, is

(A) 0 (B) 1 (C) 2 (D) 3

Q133. JEE Main 2021 (Online) 1st September Evening Shift

Consider the system of linear equations

$-x + y + 2z = 0$

$3x - ay + 5z = 1$

$2x - 2y - az = 7$

Let S_1 be the set of all $a \in R$ for which the system is inconsistent and S_2 be the set of all $a \in R$ for which the system has infinitely many solutions. If $n(S_1)$ and $n(S_2)$ denote the number of elements in S_1 and S_2 respectively, then

(A) $n(S_1) = 2, n(S_2) = 2$

(B) $n(S_1) = 1, n(S_2) = 0$

(C) $n(S_1) = 2, n(S_2) = 0$

(D) $n(S_1) = 0, n(S_2) = 2$

Q134. JEE Main 2021 (Online) 31st August Evening Shift

If $\alpha + \beta + \gamma = 2\pi$, then the system of equations

$x + (\cos\gamma)y + (\cos\beta)z = 0$

$(\cos\gamma)x + y + (\cos\alpha)z = 0$

$(\cos\beta)x + (\cos\alpha)y + z = 0$

has :

(A) no solution

(B) infinitely many solution

(C) exactly two solutions

(D) a unique solution

Q135. JEE Main 2021 (Online) 31st August Morning Shift

If the following system of linear equations

$2x + y + z = 5$

$x - y + z = 3$

$x + y + az = b$

has no solution, then :

(A) $a = -\frac{1}{3}, b \neq \frac{7}{3}$

(B) $a \neq \frac{1}{3}, b = \frac{7}{3}$

(C) $a \neq -\frac{1}{3}, b = \frac{7}{3}$

(D) $a = \frac{1}{3}, b \neq \frac{7}{3}$

Q136. JEE Main 2021 (Online) 31st August Morning Shift

If $a_r = \cos\frac{2r\pi}{9} + i\sin\frac{2r\pi}{9}$, r = 1, 2, 3, ..., i = $\sqrt{-1}$, then the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is equal to :

(A) $a_2a_6 - a_4a_8$ (B) a_9 (C) $a_1a_9 - a_3a_7$ (D) a_5

Q137. JEE Main 2021 (Online) 27th August Evening Shift

Let $A = \begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, where [t] denotes the greatest integer less than or equal to t. If det(A) = 192, then the set of values of x is the interval :

(A) [68, 69) (B) [62, 63) (C) [65, 66) (D) [60, 61)

Q138. JEE Main 2021 (Online) 27th August Evening Shift

Let $[\lambda]$ be the greatest integer less than or equal to λ. The set of all values of λ for which the system of linear equations

x + y + z = 4,

3x + 2y + 5z = 3,

9x + 4y + (28 + $[\lambda]$)z = $[\lambda]$ has a solution is :

(A) R (B) $(-\infty, -9) \cup (-9, \infty)$

(C) $[-9, -8)$ (D) $(-\infty, -9) \cup [-8, \infty)$

Q139. JEE Main 2021 (Online) 27th August Morning Shift

If the matrix $A = \begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is :

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) -1 (D) 1

Q140. JEE Main 2021 (Online) 26th August Evening Shift

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $A^{2025} - A^{2020}$ is equal to :

(A) $A^6 - A$ (B) A^5 (C) $A^5 - A$ (D) A^6

Q141. JEE Main 2021 (Online) 26th August Morning Shift

Let $\theta \in \left(0, \frac{\pi}{2}\right)$. If the system of linear equations

$(1+\cos^2\theta)x + \sin^2\theta y + 4\sin 3\theta z = 0$

$\cos^2\theta x + (1+\sin^2\theta)y + 4\sin 3\theta z = 0$

$\cos^2\theta x + \sin^2\theta y + (1+4\sin 3\theta)z = 0$

has a non-trivial solution, then the value of θ is :

(A) $\frac{4\pi}{9}$ (B) $\frac{7\pi}{18}$ (C) $\frac{\pi}{18}$ (D) $\frac{5\pi}{18}$

Q142. JEE Main 2021 (Online) 26th August Morning Shift

If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$, and Q = $A^T BA$, then the inverse of the matrix A Q^{2021} A^T is equal to :

(A) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

Q143. JEE Main 2021 (Online) 27th July Morning Shift

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A$, $\alpha, \beta \in R$, I is a 2×2 identity matrix then $4(\alpha - \beta)$ is equal to :

(A) 5 (B) $\frac{8}{3}$ (C) 2 (D) 4

Q144. JEE Main 2021 (Online) 25th July Evening Shift

The number of distinct real roots

of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$ is :

(A) 4 (B) 1 (C) 2 (D) 3

Q145. JEE Main 2021 (Online) 25th July Evening Shift

If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is :

(A) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

Q146. JEE Main 2021 (Online) 25th July Morning Shift

The values of a and b, for which the system of equations

$2x + 3y + 6z = 8$

$x + 2y + az = 5$

$3x + 5y + 9z = b$

has no solution, are :

(A) $a = 3, b \neq 13$

(B) $a \neq 3, b \neq 13$

(C) $a \neq 3, b = 3$

(D) $a = 3, b = 13$

Q147. JEE Main 2021 (Online) 22th July Evening Shift

The values of λ and μ such that the system of equations $x + y + z = 6, 3x + 5y + 5z = 26$, $x + 2y + \lambda z = \mu$ has no solution, are :

(A) $\lambda = 3, \mu = 5$

(B) $\lambda = 3, \mu \neq 10$

(C) $\lambda \neq 2, \mu = 10$

(D) $\lambda = 2, \mu \neq 10$

Q148. JEE Main 2021 (Online) 22th July Evening Shift

Let $A = [a_{ij}]$ be a real matrix of order 3×3, such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1, 2, 3$. Then, the sum of all the entries of the matrix A^3 is equal to :

(A) 2

(B) 1

(C) 3

(D) 9

Q149. JEE Main 2021 (Online) 20th July Evening Shift

The value of k $\in$R, for which the following system of linear equations

$3x - y + 4z = 3$,

$x + 2y - 3z = -2$

$6x + 5y + kz = -3$,

has infinitely many solutions, is :

(A) 3 (B) -5 (C) 5 (D) -3

Q150. JEE Main 2021 (Online) 20th July Morning Shift

Let $A = \begin{bmatrix} 2 & 3 \\ a & 0 \end{bmatrix}$, a$\in$R be written as P + Q where P is a symmetric matrix and Q is skew symmetric matrix. If det(Q) = 9, then the modulus of the sum of all possible values of determinant of P is equal to :

(A) 36 (B) 24 (C) 45 (D) 18

Q151. JEE Main 2021 (Online) 18th March Evening Shift

Let the system of linear equations

$4x + \lambda y + 2z = 0$

$2x - y + z = 0$

$\mu x + 2y + 3z = 0$, $\lambda, \mu \in$R.

has a non-trivial solution. Then which of the following is true?

(A) $\mu = 6, \lambda \in$R (B) $\lambda = 3, \mu \in$R

(C) $\mu = -6, \lambda \in$R (D) $\lambda = 2, \mu \in$R

Q152. JEE Main 2021 (Online) 18th March Morning Shift

The solutions of the equation $\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0, (0 < x < \pi)$, are

(A) $\frac{\pi}{12}, \frac{\pi}{6}$ (B) $\frac{\pi}{6}, \frac{5\pi}{6}$

(C) $\frac{5\pi}{12}, \frac{7\pi}{12}$ (D) $\frac{7\pi}{12}, \frac{11\pi}{12}$

Q153. JEE Main 2021 (Online) 18th March Morning Shift

Let α, β, γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, (a, b, c $\in$ R and a, b $\neq$ 0). If the system of equations (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$, $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is

(A) 5 (B) 3 (C) 1 (D) 0

Q154. JEE Main 2021 (Online) 18th March Morning Shift

Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$. If Tr(A) denotes the sum of all diagonal elements of the matrix A, then Tr(A) $-$ Tr(B) has value equal to

(A) 1 (B) 2 (C) 0 (D) 3

Q155. JEE Main 2021 (Online) 17th March Evening Shift

If x, y, z are in arithmetic progression with common difference d, x $\neq$ 3d, and the determinant of the matrix $\begin{bmatrix} 3 & 4\sqrt{2} & x \\ 4 & 5\sqrt{2} & y \\ 5 & k & z \end{bmatrix}$ is zero, then the value of k^2 is :

(A) 72 (B) 12 (C) 36 (D) 6

Q156. JEE Main 2021 (Online) 17th March Morning Shift

The system of equations kx + y + z = 1, x + ky + z = k and x + y + zk = k^2 has no solution if k is equal to :

(A) 0 (B) -1 (C) -2 (D) 1

Q157. JEE Main 2021 (Online) 16th March Evening Shift

The maximum value of

$$f(x) = \begin{vmatrix} \sin^2 x & 1+\cos^2 x & \cos 2x \\ 1+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}, x \in R \text{ is :}$$

(A) $\sqrt{5}$ (B) $\frac{3}{4}$ (C) 5 (D) $\sqrt{7}$

Q158. JEE Main 2021 (Online) 16th March Morning Shift

Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}, i = \sqrt{-1}$. Then, the system of linear equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has :

(A) Exactly two solutions (B) Infinitely many solutions

(C) A unique solution (D) No solution

Q159. JEE Main 2021 (Online) 26th February Evening Shift

Consider the following system of equations :

$x + 2y - 3z = a$

$2x + 6y - 11z = b$

$x - 2y + 7z = c,$

where a, b and c are real constants. Then the system of equations :

(A) has no solution for all a, b and c

(B) has a unique solution when $5a = 2b + c$

(C) has infinite number of solutions when $5a = 2b + c$

(D) has a unique solution for all a, b and c

Q160. JEE Main 2021 (Online) 26th February Morning Shift

Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

(A) 6 (B) 4 (C) 1 (D) 12

Q161. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is :

(A) -2 (B) 0

(C) (a + 2)(a + 3)(a + 4) (D) (a + 1)(a + 2)(a + 3)

Q162. JEE Main 2021 (Online) 25th February Evening Shift

Let A be a 3×3 matrix with det(A) = 4. Let R_i denote the ith row of A. If a matrix B is obtained by performing the operation $R_2 \to 2R_2 + 5R_3$ on 2A, then det(B) is equal to :

(A) 64 (B) 16 (C) 128 (D) 80

Q163. JEE Main 2021 (Online) 25th February Evening Shift

If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is :

(A) 3 (B) 2 (C) 1 (D) 4

Q164. JEE Main 2021 (Online) 25th February Evening Shift

The following system of linear equations

2x + 3y + 2z = 9

3x + 2y + 2z = 9

x − y + 4z = 8

(A) does not have any solution

(B) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

(C) has a unique solution

(D) has infinitely many solutions

Q165. JEE Main 2021 (Online) 24th February Evening Shift

Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)$ X = O, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has :

(A) no solution

(B) exactly two solutions

(C) infinitely many solutions

(D) a unique solution

Q166. JEE Main 2021 (Online) 24th February Evening Shift

For the system of linear equations:

$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R,$

consider the following statements :

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if k = -2

(C) The system has unique solution if k = 2

(D) The system has no solution if k = 2

(E) The system has infinite number of solutions if k $\neq -2$.

Which of the following statements are correct?

(A) (B) and (E) only

(B) (C) and (D) only

(C) (A) and (E) only

(D) (A) and (D) only

Q167. JEE Main 2021 (Online) 24th February Morning Shift

The system of linear equations
3x - 2y - kz = 10
2x - 4y - 2z = 6
x+2y - z = 5m
is inconsistent if :

(A) $k \neq 3, m \in \mathbf{R}$ (B) $k = 3, m \neq \frac{4}{5}$
(C) $k = 3, m = \frac{4}{5}$ (D) $k \neq 3, m \neq \frac{4}{5}$

Q168. JEE Main 2020 (Online) 6th September Evening Slot

Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$.

If $B = A + A^4$, then det (B) :

(A) lies in (1, 2) (B) lies in (2, 3). (C) is zero. (D) is one.

Q169. JEE Main 2020 (Online) 6th September Morning Slot

The values of λ and μ for which the system of linear equations
x + y + z = 2
x + 2y + 3z = 5
x + 3y + λz = μ
has infinitely many solutions are, respectively:

(A) 6 and 8 (B) 5 and 8 (C) 5 and 7 (D) 4 and 9

Q170. JEE Main 2020 (Online) 6th September Morning Slot

Let m and M be respectively the minimum and maximum values of

$$\begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x \end{vmatrix}$$

Then the ordered pair (m, M) is equal to :

(A) (−3, −1) (B) (−4, −1) (C) (1, 3) (D) (−3, 3)

Q171. JEE Main 2020 (Online) 5th September Evening Slot

If a + x = b + y = c + z + 1, where a, b, c, x, y, z are non-zero distinct real numbers, then $\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is equal to :

(A) y(b – a) (B) y(a – b) (C) y(a – c) (D) 0

Q172. JEE Main 2020 (Online) 5th September Evening Slot

If the system of linear equations

x + y + 3z = 0

x + 3y + k^2z = 0

3x + y + 3z = 0

has a non-zero solution (x, y, z) for some k $\in$ R, then x + $\left(\frac{y}{z}\right)$ is equal to :

(A) 9 (B) 3 (C) -9 (D) -3

Q173. JEE Main 2020 (Online) 5th September Morning Slot

If the minimum and the maximum values of the function $f : \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \to R$, defined by $f(\theta) = \begin{vmatrix} -\sin^2\theta & -1-\sin^2\theta & 1 \\ -\cos^2\theta & -1-\cos^2\theta & 1 \\ 12 & 10 & -2 \end{vmatrix}$ are m and M respectively, then the ordered pair (m,M) is equal to :

(A) $\left(0, 2\sqrt{2}\right)$ (B) (-4, 0) (C) (-4, 4) (D) (0, 4)

Q174. JEE Main 2020 (Online) 5th September Morning Slot

Let $\lambda \in$ R . The system of linear equations

$2x_1$ - $4x_2$ + λx_3 = 1

x_1 - $6x_2$ + x_3 = 2

λx_1 - $10x_2$ + $4x_3$ = 3

is inconsistent for:

(A) exactly one positive value of λ (B) exactly one negative value of λ

(C) exactly two values of λ (D) every value of λ

Q175. JEE Main 2020 (Online) 4th September Evening Slot

Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations,
Ax = b when the vector b on the right side is equal to b_1, b_2 and b_3 respectively. if

$$x_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, x_2 = \begin{bmatrix}0\\2\\1\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$b_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, b_2 = \begin{bmatrix}0\\2\\0\end{bmatrix} \text{ and } b_3 = \begin{bmatrix}0\\0\\2\end{bmatrix},$$

then the determinant of A is equal to :

(A) $\frac{3}{2}$ (B) 4 (C) 2 (D) $\frac{1}{2}$

Q176. JEE Main 2020 (Online) 4th September Evening Slot

If the system of equations
x+y+z=2
2x+4y−z=6
3x+2y+λz=μ
has infinitely many solutions, then

(A) $2\lambda - \mu = 5$ (B) $\lambda - 2\mu = -5$ (C) $2\lambda + \mu = 14$ (D) $\lambda + 2\mu = 14$

Q177. JEE Main 2020 (Online) 4th September Morning Slot

If $A = \begin{bmatrix}\cos\theta & i\sin\theta\\ i\sin\theta & \cos\theta\end{bmatrix}$, $\left(\theta = \frac{\pi}{24}\right)$

and $A^5 = \begin{bmatrix}a & b\\ c & d\end{bmatrix}$, where $i = \sqrt{-1}$ then which one of the following is not true?

(A) $a^2 - c^2 = 1$ (B) $0 \le a^2 + b^2 \le 1$ (C) $a^2 - d^2 = 0$ (D) $a^2 - b^2 = \frac{1}{2}$

Q178. JEE Main 2020 (Online) 3rd September Morning Slot

If $\Delta = \begin{vmatrix}x-2 & 2x-3 & 3x-4\\ 2x-3 & 3x-4 & 4x-5\\ 3x-5 & 5x-8 & 10x-17\end{vmatrix}$ =

$Ax^3 + Bx^2 + Cx + D$, then B + C is equal to :

(A) -1 (B) -3 (C) 9 (D) 1

Q179. JEE Main 2020 (Online) 2nd September Evening Slot

Let a, b, c $\in$ R be all non-zero and satisfy
$a^3 + b^3 + c^3 = 2$. If the matrix

$$A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$$

satisfies $A^TA = I$, then a value of **abc** can be :

(A) 3 (B) $\frac{1}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

Q180. JEE Main 2020 (Online) 2nd September Evening Slot

Let $A = \{X = (x, y, z)^T : PX = 0$ and

$x^2 + y^2 + z^2 = 1\}$ where

$$P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix},$$

then the set A :

(A) is an empty set.
(B) contains more than two elements.
(C) contains exactly two elements.
(D) is a singleton.

Q181. JEE Main 2020 (Online) 2nd September Morning Slot

Let S be the set of all $\lambda \in$ R for which the system of linear equations

$2x - y + 2z = 2$
$x - 2y + \lambda z = -4$
$x + \lambda y + z = 4$

has no solution. Then the set S :

(A) contains more than two elements.
(B) contains exactly two elements.
(C) is a singleton.
(D) is an empty set.

Q182. JEE Main 2020 (Online) 2nd September Morning Slot

Let A be a 2×2 real matrix with entries from {0, 1} and $|A| \neq 0$. Consider the following two statements :

(P) If $A \neq I_2$, then $|A| = -1$
(Q) If $|A| = 1$, then $tr(A) = 2$,

where I_2 denotes 2×2 identity matrix and tr(A) denotes the sum of the diagonal entries of A. Then :

(A) (P) is true and (Q) is false
(B) Both (P) and (Q) are false
(C) Both (P) and (Q) are true
(D) (P) is false and (Q) is true

Q183. JEE Main 2020 (Online) 9th January Evening Slot

The following system of linear equations
$7x + 6y - 2z = 0$
$3x + 4y + 2z = 0$
$x - 2y - 6z = 0$, has

(A) no solution
(B) infinitely many solutions, (x, y, z) satisfying $y = 2z$
(C) infinitely many solutions, (x, y, z) satisfying $x = 2z$
(D) only the trivial solution

Q184. JEE Main 2020 (Online) 9th January Evening Slot

Let $a - 2b + c = 1$.

If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then:

(A) f(50) = 1
(B) f(−50) = −1
(C) f(50) = −501
(D) f(−50) = 501

Q185. JEE Main 2020 (Online) 9th January Morning Slot

If for some α and β in R, the intersection of the following three planes
$x + 4y - 2z = 1$
$x + 7y - 5z = \beta$
$x + 5y + \alpha z = 5$
is a line in R^3, then $\alpha + \beta$ is equal to :

(A) -10
(B) 0
(C) 10
(D) 2

Q186. JEE Main 2020 (Online) 8th January Evening Slot

The system of linear equations
$\lambda x + 2y + 2z = 5$
$2\lambda x + 3y + 5z = 8$
$4x + \lambda y + 6z = 10$ has

(A) a unique solution when $\lambda = -8$
(B) no solution when $\lambda = 2$
(C) infinitely many solutions when $\lambda = 2$
(D) no solution when $\lambda = 8$

Q187. JEE Main 2020 (Online) 8th January Evening Slot

If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ then $10A^{-1}$ is equal to :

(A) 6I − A
(B) 4I − A
(C) A − 6I
(D) A − 4I

Q188. JEE Main 2020 (Online) 8th January Morning Slot

For which of the following ordered pairs (μ, δ), the system of linear equations
$x + 2y + 3z = 1$
$3x + 4y + 5z = \mu$
$4x + 4y + 4z = \delta$
is inconsistent ?

(A) (1, 0)
(B) (4, 3)
(C) (4, 6)
(D) (3, 4)

Q189. JEE Main 2020 (Online) 7th January Morning Slot

Let α be a root of the equation $x^2 + x + 1 = 0$ and the
matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$

then the matrix A^{31} is equal to

(A) A^2
(B) A
(C) I_3
(D) A^3

Q190. JEE Main 2020 (Online) 7th January Morning Slot

If the system of linear equations
$2x + 2ay + az = 0$
$2x + 3by + bz = 0$
$2x + 4cy + cz = 0$,
where a, b, c $\in$ R are non-zero distinct; has a non-zero solution, then:

(A) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.
(B) a + b + c = 0
(C) a, b, c are in G.P.
(D) a,b,c are in A.P.

Q191. JEE Main 2019 (Online) 12th April Evening Slot

A value of $\theta \in \left(0, \frac{\pi}{3}\right)$, for which $\begin{vmatrix} 1+\cos^2\theta & \sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & 1+\sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & \sin^2\theta & 1+4\cos 6\theta \end{vmatrix} = 0$, is :

(A) $\frac{\pi}{18}$ (B) $\frac{\pi}{9}$ (C) $\frac{7\pi}{24}$ (D) $\frac{7\pi}{36}$

Q192. JEE Main 2019 (Online) 12th April Morning Slot

If A is a symmetric matrix and B is a skew-symmetric matrix such that A + B = $\begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

(A) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

(B) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

Q193. JEE Main 2019 (Online) 12th April Morning Slot

If $B = \begin{bmatrix} 5 & 2\alpha & 1 \\ 0 & 2 & 1 \\ \alpha & 3 & -1 \end{bmatrix}$ is the inverse of a 3 × 3 matrix A, then the sum of all values of α for which det(A) + 1 = 0, is :

(A) 2 (B) - 1 (C) 0 (D) 1

Q194. JEE Main 2019 (Online) 10th April Evening Slot

Let λ be a real number for which the system of linear equations x + y + z = 6, 4x + λy − λz = λ − 2, 3x + 2y − 4z = − 5 has infinitely many solutions. Then λ is a root of the quadratic equation:

(A) λ^2 + λ - 6 = 0 (B) λ^2 - λ - 6 = 0

(C) λ^2 - 3λ - 4 = 0 (D) λ^2 + 3λ - 4 = 0

Q195. JEE Main 2019 (Online) 10th April Evening Slot

The sum of the real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, is equal to :

(A) - 4 (B) 0 (C) 1 (D) 6

Q196. JEE Main 2019 (Online) 10th April Morning Slot

If $\Delta_1 = \begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ and

$\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}, x \neq 0$;

then for all $\theta \in \left(0, \frac{\pi}{2}\right)$:

(A) $\Delta_1 - \Delta_2$ = x (cos 2θ − cos 4θ)

(B) $\Delta_1 + \Delta_2$ = - 2x^3

(C) $\Delta_1 + \Delta_2$ = − 2(x^3 + x −1)

(D) $\Delta_1 - \Delta_2$ = - 2x^3

Q197. JEE Main 2019 (Online) 10th April Morning Slot

If the system of linear equations

x + y + z = 5

x + 2y + 2z = 6

x + 3y + λz = μ, (λ, $\mu \in$ R), has infinitely many solutions, then the value of $\lambda + \mu$ is :

(A) 10 (B) 9 (C) 7 (D) 12

Q198. JEE Main 2019 (Online) 9th April Evening Slot

If the system of equations 2x + 3y − z = 0, x + ky − 2z = 0 and 2x − y + z = 0 has a non-trival solution (x, y, z), then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to :-

(A) -4 (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{4}$

Q199. JEE Main 2019 (Online) 9th April Evening Slot

The total number of matrices

$A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$

(x, y $\in$ R,x $\neq$ y) for which A^{T}A = 3I$_3$ is :-

(A) 3 (B) 4 (C) 2 (D) 6

Q200. JEE Main 2019 (Online) 9th April Morning Slot

Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in R,

$$\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$$

is equal to

(A) $y(y^2 - 1)$ (B) $y(y^2 - 3)$ (C) y^3 (D) $y^3 - 1$

Q201. JEE Main 2019 (Online) 9th April Morning Slot

If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}....\begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$,

then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is

(A) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$

Q202. JEE Main 2019 (Online) 8th April Evening Slot

If the system of linear equations

$x - 2y + kz = 1$
$2x + y + z = 2$
$3x - y - kz = 3$

has a solution (x,y,z), $z \neq 0$, then (x,y) lies on the straight line whose equation is :

(A) $4x - 3y - 4 = 0$ (B) $3x - 4y - 1 = 0$ (C) $4x - 3y - 1 = 0$ (D) $3x - 4y - 4 = 0$

Q203. JEE Main 2019 (Online) 8th April Evening Slot

Let the number 2,b,c be in an A.P. and

$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. If $\det(A) \in [2, 16]$, then c lies in the interval :

(A) [2, 3) (B) [4, 6] (C) $(2 + 2^{3/4}, 4)$ (D) $[3, 2 + 2^{3/4}]$

Q204. JEE Main 2019 (Online) 8th April Morning Slot

Let $A = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$, ($\alpha \in$ R)

such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ then a value of α is

(A) 0 (B) $\frac{\pi}{16}$ (C) $\frac{\pi}{32}$ (D) $\frac{\pi}{64}$

Q205. JEE Main 2019 (Online) 8th April Morning Slot

The greatest value of c $\in$ R for which the system of linear equations

x − cy − cz = 0

cx − y + cz = 0

cx + cy − z = 0

has a non-trivial solution, is :

(A) -1 (B) 0 (C) 1/2 (D) 2

Q206. JEE Main 2019 (Online) 12th January Evening Slot

If A = $\begin{bmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{bmatrix}$;

then for all $\theta \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$, det (A) lies in the interval :

(A) $\left(\frac{3}{2}, 3\right]$ (B) $\left(0, \frac{3}{2}\right]$ (C) $\left[\frac{5}{2}, 4\right)$ (D) $\left(1, \frac{5}{2}\right]$

Q207. JEE Main 2019 (Online) 12th January Evening Slot

The set of all values of λ for which the system of linear equations

x − 2y − 2z = λx

x + 2y + z = λy

− x − y = λz

has a non-trivial solution :

(A) is an empty set (B) contains more than two elements

(C) is a singleton (D) contains exactly two elements

Q208. JEE Main 2019 (Online) 12th January Morning Slot

An ordered pair (α, β) for which the system of linear equations

(1 + α) x + βy + z = 2

αx + (1 + β)y + z = 3

αx + βy + 2z = 2

has a unique solution, is :

(A) (−3, 1) (B) (1, −3) (C) (−4, 2) (D) (2, 4)

Q209. JEE Main 2019 (Online) 12th January Morning Slot

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$.

Then $\frac{q_{21}+q_{31}}{q_{32}}$ is equal to :

(A) 15 (B) 9 (C) 135 (D) 10

Q210. JEE Main 2019 (Online) 11th January Evening Slot

If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

$= (a + b + c)(x + a + b + c)^2$, $x \neq 0$,

then x is equal to :

(A) $-2(a + b + c)$ (B) $2(a + b + c)$ (C) abc (D) $-(a + b + c)$

Q211. JEE Main 2019 (Online) 11th January Morning Slot

If the system of linear equations

$2x + 2y + 3z = a$

$3x - y + 5z = b$

$x - 3y + 2z = c$

where a, b, c are non zero real numbers, has more than one solution, then :

(A) $b - c - a = 0$ (B) $a + b + c = 0$ (C) $b - c + a = 0$ (D) $b + c - a = 0$

Q212. JEE Main 2019 (Online) 11th January Morning Slot

Let $A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$. If $AA^T = I_3$, then $|p|$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{1}{\sqrt{6}}$ (D) $\frac{1}{\sqrt{3}}$

Q213. JEE Main 2019 (Online) 10th January Evening Slot

Let $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2+1 & b \\ 1 & b & 2 \end{bmatrix}$ where $b > 0$.

Then the minimum value of $\frac{\det(A)}{b}$ is -

(A) $\sqrt{3}$ (B) $-2\sqrt{3}$ (C) $-\sqrt{3}$ (D) $2\sqrt{3}$

Q214. JEE Main 2019 (Online) 10th January Evening Slot

The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$x + 3y + 7z = 0$

$-x + 4y + 7z = 0$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0.$

has a non-trivial solution, is -

(A) two (B) one (C) four (D) three

Q215. JEE Main 2019 (Online) 10th January Morning Slot

If the system of equations

$x + y + z = 5$

$x + 2y + 3z = 9$

$x + 3y + \alpha z = \beta$

has infinitely many solutions, then $\beta - \alpha$ equals -

(A) 8 (B) 21 (C) 18 (D) 5

Q216. JEE Main 2019 (Online) 10th January Morning Slot

Let $d \in R$, and

$$A = \begin{bmatrix} -2 & 4+d & (\sin\theta)-2 \\ 1 & (\sin\theta)+2 & d \\ 5 & (2\sin\theta)-d & (-\sin\theta)+2+2d \end{bmatrix},$$

$\theta \in [0, 2\pi]$ If the minimum value of det(A) is 8, then a value of d is -

(A) -7 (B) $2\left(\sqrt{2}+2\right)$ (C) -5 (D) $2\left(\sqrt{2}+1\right)$

Q217. JEE Main 2019 (Online) 9th January Evening Slot

If the system of linear equations

$x - 4y + 7z = g$

$3y - 5z = h$

$-2x + 5y - 9z = k$

is consistent, then :

(A) $g + 2h + k = 0$ (B) $g + h + 2k = 0$ (C) $2g + h + k = 0$ (D) $g + h + k = 0$

Q218. JEE Main 2019 (Online) 9th January Evening Slot

If $A = \begin{bmatrix} e^t & e^{-t}\cos t & e^{-t}\sin t \\ e^t & -e^{-t}\cos t - e^{-t}\sin t & -e^{-t}\sin t + e^{-t}cost \\ e^t & 2e^{-t}\sin t & -2e^{-t}\cos t \end{bmatrix}$

then A is :

(A) invertible for all t∈**R**.

(B) invertible only if t = π

(C) not invertible for any t∈**R**

(D) invertible only if t = $\frac{\pi}{2}$.

Q219. JEE Main 2019 (Online) 9th January Morning Slot

If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$, is equal to :

(A) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

(B) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{-1}{2} \end{bmatrix}$

(C) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

(D) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

Q220. JEE Main 2019 (Online) 9th January Morning Slot

The system of linear equations

x + y + z = 2

2x + 3y + 2z = 5

2x + 3y + (a^2 – 1) z = a + 1 then

(A) has infinitely many solutions for a = 4

(B) has a unique solution for |a| = $\sqrt{3}$

(C) is inconsistent when |a| = $\sqrt{3}$

(D) is inconsistent when a = 4

Q221. JEE Main 2018 (Online) 16th April Morning Slot

Let A = $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and B = A^{20}. Then the sum of the elements of the first column of B is :

(A) 210 (B) 211 (C) 231 (D) 251

Q222. JEE Main 2018 (Online) 16th April Morning Slot

The number of values of k for which the system of linear equations,
(k + 2)x + 10y = k
kx + (k +3)y = k -1
has **no solution,** is :

(A) 1 (B) 2 (C) 3 (D) infinitely many

Q223. JEE Main 2018 (Offline)

If the system of linear equations

x + ky + 3z = 0
3x + ky - 2z = 0
2x + 4y - 3z = 0

has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to

(A) 30 (B) -10 (C) 10 (D) -30

Q224. JEE Main 2018 (Offline)

If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$

then the ordered pair (A, B) is equal to :

(A) (4, 5) (B) (-4, -5) (C) (-4, 3) (D) (-4, 5)

Q225. JEE Main 2018 (Online) 15th April Evening Slot

Suppose A is any 3×3 non-singular matrix and $(A-3I)(A-5I) = O$ where $I = I_3$ and $O = O_3$. If $\alpha A + \beta A^{-1} = 4I$, then $\alpha + \beta$ is equal to :

(A) 8 (B) 7 (C) 13 (D) 12

Q226. JEE Main 2018 (Online) 15th April Evening Slot

If the system of linear equations
x + ay + z = 3
x + 2y + 2z = 6
x + 5y + 3z = b
has no solution, then :

(A) $a = -1,\ b = 9$ (B) $a = -1,\ b \neq 9$
(C) $a \neq -1,\ b = 9$ (D) $a = 1,\ b \neq 9$

Q227. JEE Main 2018 (Online) 15th April Morning Slot

Let A be a matrix such that $A.\begin{bmatrix}1 & 2\\0 & 3\end{bmatrix}$ is a scalar matrix and |3A| = 108.

Then A^2 equals :

(A) $\begin{bmatrix}4 & -32\\0 & 36\end{bmatrix}$

(B) $\begin{bmatrix}36 & 0\\-32 & 4\end{bmatrix}$

(C) $\begin{bmatrix}4 & 0\\-32 & 36\end{bmatrix}$

(D) $\begin{bmatrix}36 & -32\\0 & 4\end{bmatrix}$

Q228. JEE Main 2018 (Online) 15th April Morning Slot

Let S be the set of all real values of k for which the systemof linear equations

x + y + z = 2

2x + y − z = 3

3x + 2y + kz = 4

has a unique solution. Then S is :

(A) an empty set (B) equal to {0} (C) equal to **R** (D) equal to **R** − {0}

Q229. JEE Main 2017 (Online) 9th April Morning Slot

For two 3 × 3 matrices A and B, let $A + B = 2B^T$ and $3A + 2B = I_3$, where B^T is the transpose of B and I_3 is 3 × 3 identity matrix. Then :

(A) $5A + 10B = 2I_3$ (B) $10A + 5B = 3I_3$ (C) $B + 2A = I_3$ (D) $3A + 6B = 2I_3$

Q230. JEE Main 2017 (Online) 8th April Morning Slot

The number of real values of λ for which the system of linear equations

$2x + 4y - \lambda z = 0$

$4x + \lambda y + 2z = 0$

$\lambda x + 2y + 2z = 0$

has infinitely many solutions, is :

(A) 0 (B) 1 (C) 2 (D) 3

Q231. JEE Main 2017 (Online) 8th April Morning Slot

Let A be any 3 × 3 invertible matrix. Then which one of the following is **not** always true ?

(A) adj (A) = |A|.A^{-1}

(B) adj (adj(A)) = |A|.A

(C) adj (adj(A)) = $|A|^2$.$(adj(A))^{-1}$

(D) adj (adj(A)) = | A | .$(adj(A))^{-1}$

Q232. JEE Main 2017 (Online) 8th April Morning Slot

If

$$S = \left\{ x \in [0, 2\pi] : \begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix} = 0 \right\},$$

then $\sum_{x \in S} \tan\left(\frac{\pi}{3} + x\right)$ is equal to :

(A) $4 + 2\sqrt{3}$ (B) $-2 + \sqrt{3}$ (C) $-2 - \sqrt{3}$ (D) $-4 - 2\sqrt{3}$

Q233. JEE Main 2017 (Offline)

If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$,

then adj($3A^2$ + 12A) is equal to

(A) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

(B) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$

(C) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

(D) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$

Q234. JEE Main 2017 (Offline)

If S is the set of distinct values of 'b' for which the following system of linear equations

$x + y + z = 1$
$x + ay + z = 1$
$ax + by + z = 0$

has no solution, then S is :

(A) an empty set

(B) an infinite set

(C) a finite set containing two or more elements

(D) a singleton

Q235. JEE Main 2016 (Online) 10th April Morning Slot

Let A be a 3×3 matrix such that $A^2 - 5A + 7I = 0$

Statement - I :

$A^{-1} = \frac{1}{7}(5I - A)$.

Statement - II :

The polynomial $A^3 - 2A^2 - 3A + I$ can be reduced to $5(A - 4I)$.

Then :

(A) Statement-I is true, but Statement-II is false.

(B) Statement-I is false, but Statement-II is true.

(C) Both the statements are true.

(D) Both the statements are false

Q236. JEE Main 2016 (Online) 10th April Morning Slot

If $A = \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix}$,

then the determinant of the matrix $(A^{2016} - 2A^{2015} - A^{2014})$ is :

(A) 2014 (B) -175 (C) 2016 (D) -25

Q237. JEE Main 2016 (Online) 9th April Morning Slot

If P = $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Q = PAP^T, then P^T Q^{2015} P is :

(A) $\begin{bmatrix} 0 & 2015 \\ 0 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 2015 & 1 \\ 0 & 2015 \end{bmatrix}$

(C) $\begin{bmatrix} 2015 & 0 \\ 1 & 2015 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 2015 \\ 0 & 1 \end{bmatrix}$

Q238. JEE Main 2016 (Online) 9th April Morning Slot

The number of distinct real roots of the equation,

$\begin{vmatrix} \cos x & \sin x & \sin x \\ \sin x & \cos x & \sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = 0$ in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is :

(A) 4 (B) 3 (C) 2 (D) 1

Q239. JEE Main 2016 (Offline)

If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and A adj $A = A\,A^T$, then $5a + b$ is equal to :

(A) 4 (B) 13 (C) -1 (D) 5

Q240. JEE Main 2016 (Offline)

The system of linear equations

$x + \lambda y - z = 0$

$\lambda x - y - z = 0$

$x + y - \lambda z = 0$

has a non-trivial solution for :

(A) infinitely many values of λ.

(B) exactly one value of λ.

(C) exactly two values of λ.

(D) exactly three values of λ.

Q241. JEE Main 2015 (Offline)

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation

$AA^T = 9$I, where I is 3×3 identity matrix, then the ordered

pair (a, b) is equal to :

(A) $(2, 1)$ (B) $(-2, -1)$ (C) $(2, -1)$ (D) $(-2, 1)$

Q242. JEE Main 2015 (Offline)

The set of all values of λ for which the system of linear equations:

$$\begin{aligned} 2x_1 - 2x_2 + x_3 &= \lambda x_1 \\ 2x_1 - 3x_2 + 2x_3 &= \lambda x_2 \\ -x_1 + 2x_2 &= \lambda x_3 \end{aligned}$$

has a non-trivial solution

(A) contains two elements

(B) contains more than two elements

(C) in an empty set

(D) is a singleton

Q243. JEE Main 2014 (Offline)

If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and

$$\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix}$$

$= K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then K is equal to :

(A) 1 (B) -1 (C) $\alpha\beta$ (D) $\frac{1}{\alpha\beta}$

Q244. JEE Main 2014 (Offline)

If A is a 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, then BB' equals:

(A) B^{-1} (B) $\left(B^{-1}\right)'$ (C) $I + B$ (D) I

Q245. JEE Main 2013 (Offline)

The number of values of k, for which the system of equations :

$$\begin{aligned} (k+1)x + 8y &= 4k \\ kx + (k+3)y &= 3k - 1 \end{aligned}$$

has no solution, is

(A) infinite (B) 1 (C) 2 (D) 3

Q246. AIEEE 2011

Let A and B be two symmetric matrices of order 3.

Statement - 1 : $A(BA)$ and $(AB)A$ are symmetric matrices.

Statement - 2 : AB is symmetric matrix if matrix multiplication of A with B is commutative.

(A) statement - 1 is true, statement - 2 is true; statement - 2 is **not** a correct explanation for statement - 1.

(B) statement - 1 is true, statement - 2 is false.

(C) statement - 1 is false, statement -2 is true

(D) statement -1 is true, statement - 2 is true; statement - 2 is a correct explanation for statement - 1.

Q247. AIEEE 2011

The number of values of k for which the linear equations
$4x + ky + 2z = 0, kx + 4y + z = 0$ and $2x + 2y + z = 0$ possess a non-zero solution is :

(A) 2 (B) 1 (C) zero (D) 3

Q248. AIEEE 2010

Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$,
where I is 2×2 identity matrix. Define
$Tr(A) =$ sum of diagonal elements of A and $|A| =$ determinant of matrix A.
Statement- 1: $Tr(A) = 0$.
Statement- 2: $|A| = 1$.

(A) statement - 1 is true, statement - 2 is true; statement - 2 is **not** a correct explanation for statement - 1.

(B) statement - 1 is true, statement - 2 is false.

(C) statement - 1 is false, statement -2 is true

(D) statement -1 is true, statement - 2 is true; statement - 2 is a correct explanation for statement - 1.

Q249. AIEEE 2010

Consider the system of linear equations; $\begin{aligned} x_1 + 2x_2 + x_3 &= 3 \\ 2x_1 + 3x_2 + x_3 &= 3 \\ 3x_1 + 5x_2 + 2x_3 &= 1 \end{aligned}$

The system has :

(A) exactly 3 solutions

(B) a unique solution

(C) no solution

(D) infinite number of solutions

Q250. AIEEE 2010

The number of 3×3 non-singular matrices, with four entries as 1 and all other entries as 0, is :

(A) 5 (B) 6 (C) at least 7 (D) less than 4

Q251. AIEEE 2009

Let a, b, c be such that $b\,(a+c) \neq 0$ if

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^{n}c \end{vmatrix} = 0$$

then the value of n :

(A) any even integer (B) any odd integer (C) any integer (D) zero

Q252. AIEEE 2008

Let A be a 2×2 matrix with real entries. Let I be the 2×2 identity matrix. Denote by tr(A), the sum of diagonal entries of A. Assume that $A^2 = I$.

Statement-1 : If $A \neq I$ and $A \neq -I$, then det$(A) = -1$

Statement- 2 : If $A \neq I$ and $A \neq -I$, then tr $(A) \neq 0$.

(A) statement - 1 is false, statement -2 is true

(B) statement -1 is true, statement - 2 is true; statement - 2 is a correct explanation for statement - 1.

(C) statement - 1 is true, statement - 2 is true; statement - 2 is not a correct explanation for statement - 1.

(D) statement - 1 is true, statement - 2 is false.

Q253. AIEEE 2008

Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx$, and $z = bx + ay$. Then $a^2 + b^2 + c^2 + 2abc$ is equal to :

(A) 2 (B) -1 (C) 0 (D) 1

Q254. AIEEE 2007

If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$, then D is :

(A) divisible by x but not y (B) divisible by y but not x

(C) divisible by neither x nor y (D) divisible by both x and y

Q255. AIEEE 2006

If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true?

(A) $A = B$

(B) $AB = BA$

(C) either of A or B is a zero matrix

(D) either of A or B is identity matrix

Q256. AIEEE 2006

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, a, b \in N$. Then

(A) there cannot exist any B such that $AB = BA$

(B) there exist more then one but finite number of B's such that $AB = BA$

(C) there exists exactly one B such that $AB = BA$

(D) there exist infinitely many B's such that $AB = BA$

Q257. AIEEE 2005

If $A^2 - A + I = 0$, then the inverse of A is :

(A) $A + I$

(B) A

(C) $A - I$

(D) $I - A$

Q258. AIEEE 2005

The system of equations
$\alpha x + y + z = \alpha - 1$
$x + \alpha y + z = \alpha - 1$
$x + y + \alpha z = \alpha - 1$
has no solutions, if α is :

(A) -2

(B) either -2 or 1

(C) not -2

(D) 1

Q259. AIEEE 2005

If $a^2 + b^2 + c^2 = -2$ and
$\mathrm{f}(x) = \begin{vmatrix} 1 + a^2x & (1 + b^2)x & (1 + c^2)x \\ (1 + a^2)x & 1 + b^2x & (1 + c^2)x \\ (1 + a^2)x & (1 + b^2)x & 1 + c^2x \end{vmatrix}$,
then $\mathrm{f}(x)$ is a polynomial of degree :

(A) 1

(B) 0

(C) 3

(D) 2

Q260. AIEEE 2005

If $a_1, a_2, a_3, \ldots\ldots\ldots, a_n, \ldots\ldots$ are in G.P., then the determinant $\Delta = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ is equal to :

(A) 1 (B) 0 (C) 4 (D) 2

Q261. AIEEE 2004

Let $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. The only correct statement about the matrix A is

(A) $A^2 = I$ (B) $A = (-1)I$, where I is a unit matrix

(C) A^{-1} does not exist (D) A is a zero matrix

Q262. AIEEE 2004

Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and $10B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$. If B is the inverse of matrix A, then α is

(A) 5 (B) -1 (C) 2 (D) -2

Q263. AIEEE 2004

If $a_1, a_2, a_3, \ldots\ldots\ldots, a_n, \ldots\ldots$ are in G.P., then the value of the determinant $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$, is

(A) -2 (B) 1 (C) 2 (D) 0

Q264. AIEEE 2003

If the system of linear equations

$x + 2ay + az = 0;\ x + 3by + bz = 0;\ x + 4cy + cz = 0;$

has a non - zero solution, then a, b, c.

(A) satisfy $a + 2b + 3c = 0$ (B) are in A.P

(C) are in G.P (D) are in H.P.

Q265. AIEEE 2003

If $1, \omega, \omega^2$ are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to

(A) ω^2 (B) 0 (C) 1 (D) ω

Q266. AIEEE 2003

If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

(A) $\alpha = 2ab,\ \beta = a^2 + b^2$

(B) $\alpha = a^2 + b^2,\ \beta = ab$

(C) $\alpha = a^2 + b^2,\ \beta = 2ab$

(D) $\alpha = a^2 + b^2,\ \beta = a^2 - b^2$

Q267. AIEEE 2002

If $a > 0$ and discriminant of $ax^2 + 2bx + c$ is $-ve$, then $\begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ ax+b & bx+c & 0 \end{vmatrix}$ is equal to

(A) $+ve$

(B) $\left(ac - b^2\right)\left(ax^2 + 2bx + c\right)$

(C) $-ve$

(D) 0

Numerical 71 questions

Q268. JEE Main 2026 (Online) 6th April Morning Shift

Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy

$\mathrm{A}^2 + \alpha(\mathrm{adj}(\mathrm{adj}(\mathrm{A}))) + \beta(\mathrm{adj}(\mathrm{A})(\mathrm{adj}(\mathrm{adj}(\mathrm{A})))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$.

Then $(\alpha - \beta)^2$ is equal to ____

Q269. JEE Main 2026 (Online) 2nd April Evening Shift

Consider the matrices $A = \begin{bmatrix} 2 & -2 \\ 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$. If matrices P and Q are such that $PA = B$ and $AQ = B$, then the absolute value of the sum of the diagonal elements of $2(P + Q)$ is _____.

Q270. JEE Main 2026 (Online) 28th January Evening Shift

Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is _____

Q271. JEE Main 2026 (Online) 24th January Morning Shift

The number of 3×2 matrices A , which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of $\mathrm{A}^{\mathrm{T}}\mathrm{A}$ is 5 , is

____

Q272. JEE Main 2026 (Online) 23rd January Evening Shift

Let $A=\begin{bmatrix}0 & 2 & -3\\ -2 & 0 & 1\\ 3 & -1 & 0\end{bmatrix}$ and B be a matrix such that $B(I-A)=I+A$. Then the sum of the diagonal elements of $\mathrm{B}^{\mathrm{T}}\mathrm{B}$ is equal to ____

Q273. JEE Main 2026 (Online) 23rd January Morning Shift

Let $|\mathrm{A}|=6$, where A is a 3×3 matrix. If $\left|\operatorname{adj}\left(3\operatorname{adj}\left(\mathrm{A}^2\cdot\operatorname{adj}(2\,\mathrm{A})\right)\right)\right|=2^{\mathrm{m}}\cdot 3^{\mathrm{n}},\mathrm{m},\mathrm{n}\in\mathbf{N}$, then $\mathrm{m}+\mathrm{n}$ is equal to

____ .

Q274. JEE Main 2026 (Online) 22nd January Morning Shift

Let A be a 3×3 matrix such that $\mathrm{A}+\mathrm{A}^{\mathrm{T}}=\mathrm{O}$. If $\mathrm{A}\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}3\\3\\2\end{bmatrix},\mathrm{A}^2\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}-3\\19\\-24\end{bmatrix}$ and $\det(\operatorname{adj}(2\operatorname{adj}(\mathrm{A}+\mathrm{I})))=(2)^{\alpha}\cdot(3)^{\beta}\cdot(11)^{\gamma},\alpha,\beta,\gamma$ are non-negative integers, then $\alpha+\beta+\gamma$ is equal to ____

Q275. JEE Main 2026 (Online) 21st January Morning Shift

For some $\alpha,\beta\in\mathbf{R}$, let $A=\begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}1 & 1\\ 1 & \beta\end{bmatrix}$ be such that $A^2-4A+2I=B^2-3B+I=O$. Then $\left(\det\left(\operatorname{adj}\left(A^3-B^3\right)\right)\right)^2$ is equal to ____ .

Q276. JEE Main 2025 (Online) 7th April Morning Shift

The number of singular matrices of order 2 , whose elements are from the set $\{2,3,6,9\}$, is _____.

Q277. JEE Main 2025 (Online) 4th April Morning Shift

Let $A=\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\ 0 & 1 & 0\\ \sin\theta & 0 & \cos\theta\end{bmatrix}$. If for some $\theta\in(0,\pi),A^2=A^T$, then the sum of the diagonal elements of the matrix $(\mathrm{A}+\mathrm{I})^3+(\mathrm{A}-\mathrm{I})^3-6\,\mathrm{A}$ is equal to _____ .

Q278. JEE Main 2025 (Online) 3rd April Evening Shift

Let I be the identity matrix of order 3×3 and for the matrix $A=\begin{bmatrix}\lambda & 2 & 3\\ 4 & 5 & 6\\ 7 & -1 & 2\end{bmatrix},|A|=-1$. Let B be the inverse of the matrix $\operatorname{adj}\left(A\operatorname{adj}\left(A^2\right)\right)$. Then $|(\lambda\mathrm{B}+\mathrm{I})|$ is equal to_____

Q279. JEE Main 2025 (Online) 29th January Morning Shift

Let $S=\left\{m \in \mathbf{Z}: A^{m^2}+A^m=3I-A^{-6}\right\}$, where $A=\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix}$. Then $n(S)$ is equal to _____.

Q280. JEE Main 2025 (Online) 28th January Morning Shift

Let M denote the set of all real matrices of order 3×3 and let $\mathrm{S}=\{-3,-2,-1,1,2\}$. Let
$\mathrm{S}_1=\left\{\mathrm{A}=\left[a_{\mathrm{ij}}\right] \in \mathrm{M}: \mathrm{A}=\mathrm{A}^{\mathrm{T}} \text{ and } a_{\mathrm{ij}} \in \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\right\}$,
$\mathrm{S}_2=\left\{\mathrm{A}=\left[a_{\mathrm{ij}}\right] \in \mathrm{M}: \mathrm{A}=-\mathrm{A}^{\mathrm{T}} \text{ and } a_{\mathrm{ij}} \in \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\right\}$,
$\mathrm{S}_3=\left\{\mathrm{A}=\left[a_{\mathrm{ij}}\right] \in \mathrm{M}: a_{11}+a_{22}+a_{33}=0 \text{ and } a_{\mathrm{ij}} \in \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\right\}$.
If $n\left(S_1 \cup S_2 \cup S_3\right)=125 \alpha$, then α equals _____.

Q281. JEE Main 2025 (Online) 24th January Morning Shift

Let A be a 3×3 matrix such that $\mathrm{X}^{\mathrm{T}} \mathrm{AX}=\mathrm{O}$ for all nonzero 3×1 matrices $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$. If $\mathrm{A}\begin{bmatrix}1\\ 1\\ 1\end{bmatrix}=\begin{bmatrix}1\\ 4\\ -5\end{bmatrix}, \mathrm{A}\begin{bmatrix}1\\ 2\\ 1\end{bmatrix}=\begin{bmatrix}0\\ 4\\ -8\end{bmatrix}$, and $\det(\operatorname{adj}(2(\mathrm{A}+\mathrm{I})))=2^\alpha 3^\beta 5^\gamma, \alpha, \beta, \gamma \in N$, then $\alpha^2+\beta^2+\gamma^2$ is

Q282. JEE Main 2025 (Online) 22nd January Morning Shift

Let A be a square matrix of order 3 such that $\det(A)=-2$ and $\det(3 \operatorname{adj}(-6 \operatorname{adj}(3A)))=2^{m+n} \cdot 3^{mn}, m>n$. Then $4m+2n$ is equal to _____.

Q283. JEE Main 2024 (Online) 9th April Evening Shift

Consider the matrices : $A=\begin{bmatrix}2 & -5\\ 3 & m\end{bmatrix}, B=\begin{bmatrix}20\\ m\end{bmatrix}$ and $X=\begin{bmatrix}x\\ y\end{bmatrix}$. Let the set of all m, for which the system of equations $AX=B$ has a negative solution (i.e., $x<0$ and $y<0$), be the interval (a, b). Then $8\int_a^b |A| dm$ is equal to _____.

Q284. JEE Main 2024 (Online) 9th April Morning Shift

Let A be a non-singular matrix of order 3. If $\det(3 \operatorname{adj}(2 \operatorname{adj}((\det A) A)))=3^{-13} \cdot 2^{-10}$ and $\det(3 \operatorname{adj}(2\mathrm{A}))=2^{\mathrm{m}} \cdot 3^{\mathrm{n}}$, then $|3 \mathrm{m}+2 \mathrm{n}|$ is equal to _____.

Q285. JEE Main 2024 (Online) 8th April Morning Shift

Let $A=\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n, then n is equal to _____.

Q286. JEE Main 2024 (Online) 6th April Evening Shift

If the system of equations

$2x + 7y + \lambda z = 3$

$3x + 2y + 5z = 4$

$x + \mu y + 32z = -1$

has infinitely many solutions, then $(\lambda - \mu)$ is equal to _____ :

Q287. JEE Main 2024 (Online) 6th April Morning Shift

Let $\alpha\beta\gamma = 45; \alpha, \beta, \gamma \in \mathbb{R}$. If $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ for some $x, y, z \in \mathbb{R}, xyz \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to _____.

Q288. JEE Main 2024 (Online) 4th April Evening Shift

Let A be a 2×2 symmetric matrix such that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ and the determinant of A be 1 . If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2, then $\alpha + \beta$ equals _____.

Q289. JEE Main 2024 (Online) 4th April Morning Shift

Let A be a square matrix of order 2 such that $|A| = 2$ and the sum of its diagonal elements is -3 . If the points (x, y) satisfying $\mathrm{A}^2 + x\,\mathrm{A} + y\mathrm{I} = \mathrm{O}$ lie on a hyperbola, whose transverse axis is parallel to the x-axis, eccentricity is e and the length of the latus rectum is l, then $\mathrm{e}^4 + l^4$ is equal to _____.

Q290. JEE Main 2024 (Online) 4th April Morning Shift

Let A be a 3×3 matrix of non-negative real elements such that $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Then the maximum value of $\det(\mathrm{A})$ is _____.

Q291. JEE Main 2024 (Online) 1st February Evening Shift

Let $A = I_2 - 2MM^T$, where M is a real matrix of order 2×1 such that the relation $M^TM = I_1$ holds. If λ is a real number such that the relation $AX = \lambda X$ holds for some non-zero real matrix X of order 2×1, then the sum of squares of all possible values of λ is equal to _____.

Q292. JEE Main 2024 (Online) 31st January Evening Shift

Let A be a 3×3 matrix and $\det(A) = 2$. If $n = \det(\underbrace{\mathrm{adj}(\mathrm{adj}(\ldots..(\mathrm{adj}\, A)))}_{2024-\text{ times}})$, then the remainder when n is divided by 9 is equal to _____.

Q293. JEE Main 2024 (Online) 29th January Evening Shift

Let for any three distinct consecutive terms a, b, c of an A.P, the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ and

$x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____.

Q294. JEE Main 2024 (Online) 27th January Evening Shift

Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|\mathrm{A} - x\mathrm{I}| = 0$ be -1 and 3, then the sum of the diagonal elements of the matrix A^2 is

Q295. JEE Main 2024 (Online) 27th January Morning Shift

Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $B = [B_1, B_2, B_3]$, where B_1, B_2, B_3 are column matrices, and

$$\mathrm{AB}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathrm{AB}_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \quad \mathrm{AB}_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

If $\alpha = |B|$ and β is the sum of all the diagonal elements of B, then $\alpha^3 + \beta^3$ is equal to _____.

Q296. JEE Main 2023 (Online) 12th April Morning Shift

Let $\mathrm{D_k} = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$. If $\sum_{k=1}^{n} \mathrm{D_k} = 96$, then n is equal to _____.

Q297. JEE Main 2023 (Online) 11th April Morning Shift

Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in \mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n \in \mathbb{N}$, then n is equal to _____.

Q298. JEE Main 2023 (Online) 10th April Evening Shift

Let S be the set of values of λ, for which the system of equations

$6\lambda x - 3y + 3z = 4\lambda^2$,

$2x + 6\lambda y + 4z = 1$,

$3x + 2y + 3\lambda z = \lambda$ has no solution. Then $12 \sum_{\lambda \in S} |\lambda|$ is equal to _____.

Q299. JEE Main 2023 (Online) 29th January Evening Shift

Let A be a symmetric matrix such that $|\mathrm{A}| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s, then $\frac{\beta s}{\alpha^2}$ is equal to _____.

Q300. JEE Main 2023 (Online) 25th January Morning Shift

Let $\mathrm{A}_1, \mathrm{A}_2, \mathrm{A}_3$ be the three A.P. with the same common difference d and having their first terms as $\mathrm{A}, \mathrm{A}+1, \mathrm{A}+2$, respectively. Let a, b, c be the $7^{\text{th}}, 9^{\text{th}}, 17^{\text{th}}$ terms of $\mathrm{A}_1, \mathrm{A}_2, \mathrm{A}_3$, respectively, such that $\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$.

If $a = 29$, then the sum of first 20 terms of an AP whose first term is $c - a - b$ and common difference is $\frac{d}{12}$, is equal to _____.

Q301. JEE Main 2022 (Online) 29th July Evening Shift

Let $X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $\mathrm{k} \in N$, if $X'A^kX = 33$, then k is equal to _____.

Q302. JEE Main 2022 (Online) 29th July Morning Shift

Let p and p + 2 be prime numbers and let

$$\Delta = \begin{vmatrix} \mathrm{p}! & (\mathrm{p}+1)! & (\mathrm{p}+2)! \\ (\mathrm{p}+1)! & (\mathrm{p}+2)! & (\mathrm{p}+3)! \\ (\mathrm{p}+2)! & (\mathrm{p}+3)! & (\mathrm{p}+4)! \end{vmatrix}$$

Then the sum of the maximum values of α and β, such that p^α and $(\mathrm{p}+2)^\beta$ divide Δ, is _____.

Q303. JEE Main 2022 (Online) 28th July Morning Shift

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}, \alpha, \beta \in \mathbf{R}$. Let α_1 be the value of α which satisfies $(\mathrm{A}+\mathrm{B})^2 = \mathrm{A}^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ and α_2 be the value of α which satisfies $(\mathrm{A}+\mathrm{B})^2 = \mathrm{B}^2$. Then $|\alpha_1 - \alpha_2|$ is equal to _____.

Q304. JEE Main 2022 (Online) 27th July Morning Shift

Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices $A \in S$ such that the sum of all the diagonal elements of $A^{\mathrm{T}}A$ is 6 is _____.

Q305. JEE Main 2022 (Online) 26th July Evening Shift

The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \{-1, 0, 1, 2, 3, \ldots\ldots, 10\}$, such that $A = A^{-1}$, is _____.

Q306. JEE Main 2022 (Online) 25th July Evening Shift

Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}, a, b \in \mathbb{R}$. If for some $n \in \mathbb{N}, A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ then $n + a + b$ is equal to _____.

Q307. JEE Main 2022 (Online) 25th July Morning Shift

Let $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ and $B = A - I$. If $\omega = \frac{\sqrt{3}i - 1}{2}$, then the number of elements in the set $\{n \in \{1, 2, \ldots, 100\} : A^n + (\omega B)^n = A + B\}$ is equal to _____.

Q308. JEE Main 2022 (Online) 29th June Evening Shift

Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number an $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is _____.

Q309. JEE Main 2022 (Online) 28th June Evening Shift

If the system of linear equations
$2x - 3y = \gamma + 5$,
$\alpha x + 5y = \beta + 1$, where $\alpha, \beta, \gamma \in$ R has infinitely many solutions then the value of $| 9\alpha + 3\beta + 5\gamma |$ is equal to _____.

Q310. JEE Main 2022 (Online) 28th June Evening Shift

Let $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where $i = \sqrt{-1}$. Then, the number of elements in the set { n ∈ {1, 2, ..., 100} : A^n = A } is _____.

Q311. JEE Main 2022 (Online) 26th June Evening Shift

Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $Y = \alpha I + \beta X + \gamma X^2$ and $Z = \alpha^2 I - \alpha\beta X + (\beta^2 - \alpha\gamma)X^2$, $\alpha, \beta, \gamma \in$ R. If $Y^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$, then $(\alpha - \beta + \gamma)^2$ is equal to _____.

Q312. JEE Main 2022 (Online) 25th June Evening Shift

Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set {(n, m) : n, m $\in$ {1, 2, ..., 10} and $nA^n + mB^m = I$} is _____.

Q313. JEE Main 2022 (Online) 24th June Evening Shift

Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \ldots . 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is _____.

Q314. JEE Main 2021 (Online) 31st August Evening Shift

The number of elements in the set $\left\{ A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in \{-1, 0, 1\} \, and \, (I - A)^3 = I - A^3 \right\}$, where I is 2×2 identity matrix, is :

Q315. JEE Main 2021 (Online) 27th August Morning Shift

If the system of linear equations

$2x + y - z = 3$

$x - y - z = \alpha$

$3x + 3y + \beta z = 3$

has infinitely many solutions, then $\alpha + \beta - \alpha\beta$ is equal to _____.

Q316. JEE Main 2021 (Online) 27th July Evening Shift

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + ... + A^{20}$, then the sum of all the elements of the matrix M is equal to _____.

Q317. JEE Main 2021 (Online) 27th July Morning Shift

For real numbers α and β, consider the following system of linear equations :

x + y $-$ z = 2, x + 2y + αz = 1, 2x $-$ y + z = β. If the system has infinite solutions, then $\alpha + \beta$ is equal to _____.

Q318. JEE Main 2021 (Online) 27th July Morning Shift

Let $f(x) = \begin{vmatrix} \sin^2 x & -2+\cos^2 x & \cos 2x \\ 2+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1+\cos 2x \end{vmatrix}, x \in [0, \pi]$. Then the maximum value of f(x) is equal to _____.

Q319. JEE Main 2021 (Online) 25th July Morning Shift

Let $S = \left\{ n \in N \middle| \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right\}$, where i = $\sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____.

Q320. JEE Main 2021 (Online) 22th July Evening Shift

Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3 $\times$ 3 matrices B with entries from the set {1, 2, 3, 4, 5} and satisfying AB = BA is _____.

Q321. JEE Main 2021 (Online) 20th July Morning Shift

Let $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ and B = $7A^{20} - 20A^7 + 2I$, where I is an identity matrix of order 3 $\times$ 3. If B = $[b_{ij}]$, then b_{13} is equal to _____.

Q322. JEE Main 2021 (Online) 20th July Morning Shift

Let a, b, c, d in arithmetic progression with common difference λ. If $\begin{vmatrix} x+a-c & x+b & x+a \\ x-1 & x+c & x+b \\ x-b+d & x+d & x+c \end{vmatrix} = 2$, then value of λ^2 is equal to _____.

Q323. JEE Main 2021 (Online) 18th March Evening Shift

Let I be an identity matrix of order 2 $\times$ 2 and P = $\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of n$\in$N for which $P^n = 5I - 8P$ is equal to _____.

Q324. JEE Main 2021 (Online) 17th March Evening Shift

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that AB = B and a + d = 2021, then the value of ad $-$ bc is equal to _____.

Q325. JEE Main 2021 (Online) 17th March Evening Shift

If 1, $\log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x, then the value of the determinant $\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x - 1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ is equal to :

Q326. JEE Main 2021 (Online) 16th March Evening Shift

Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2 $\times$ 1 matrices with real entries such that A = XB, where

$X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and k$\in$R.

If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)\, b_2^2 \neq -2b_1b_2$, then the value of k is _____.

Q327. JEE Main 2021 (Online) 16th March Morning Shift

The total number of 3 $\times$ 3 matrices A having entries from the set {0, 1, 2, 3} such that the sum of all the diagonal entries of AA^T is 9, is equal to _____.

Q328. JEE Main 2021 (Online) 26th February Evening Shift

If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation

$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α and β, then $\beta - \alpha$ is equal to _____.

Q329. JEE Main 2021 (Online) 25th February Morning Shift

If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and

$(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to

Q330. JEE Main 2021 (Online) 25th February Morning Shift

If the system of equations

$kx + y + 2z = 1$

$3x - y - 2z = 2$

$-2x - 2y - 4z = 3$

has infinitely many solutions, then k is equal to _____.

Q331. JEE Main 2021 (Online) 24th February Morning Shift

Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in R$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in R$.

If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to _____.

Q332. JEE Main 2021 (Online) 24th February Morning Shift

Let M be any 3×3 matrix with entries from the set {0, 1, 2}. The maximum number of such matrices, for which the sum of diagonal elements of M^TM is seven, is _____.

Q333. JEE Main 2020 (Online) 6th September Evening Slot

The sum of distinct values of λ for which the system of equations

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$
$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$
$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

has non-zero solutions, is _____ .

Q334. JEE Main 2020 (Online) 4th September Morning Slot

If the system of equations

$x - 2y + 3z = 9$

$2x + y + z = b$

$x - 7y + az = 24,$

has infinitely many solutions, then a - b is equal to.....

Q335. JEE Main 2020 (Online) 3rd September Evening Slot

Let S be the set of all integer solutions, (x, y, z), of the system of equations

$x - 2y + 5z = 0$

$-2x + 4y + z = 0$

$-7x + 14y + 9z = 0$

such that $15 \le x^2 + y^2 + z^2 \le 150$. Then, the number of elements in the set S is equal to _____ .

Q336. JEE Main 2020 (Online) 3rd September Morning Slot

Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in R$ and $A^4 = [a_{ij}]$.

If $a_{11} = 109$, then a_{22} is equal to _____ .

Q337. JEE Main 2020 (Online) 8th January Morning Slot

The number of all 3 × 3 matrices A, with enteries from the set {−1, 0, 1} such that the sum of the diagonal elements of AA^T is 3, is

Q338. JEE Main 2020 (Online) 7th January Evening Slot

If the system of linear equations,

$x + y + z = 6$

$x + 2y + 3z = 10$

$3x + 2y + \lambda z = \mu$

has more than two solutions, then $\mu - \lambda^2$ is equal to _____.

Out of syllabus 46 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q339. JEE Main 2023 (Online) 15th April Morning Shift · **MCQ (Single Correct Answer)**

Let the determinant of a square matrix A of order m be $m - n$, where m and n satisfy $4m + n = 22$ and $17m + 4n = 93$.

If $\det(n\,\mathrm{adj}(\mathrm{adj}(mA))) = 3^a 5^b 6^c$ then $a + b + c$ is equal to :

(A) 96 (B) 84 (C) 109 (D) 101

Q340. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Let the system of linear equations

$-x+2y-9z=7$

$-x+3y+7z=9$

$-2x+y+5z=8$

$-3x+y+13z=\lambda$

has a unique solution $x=\alpha, y=\beta, z=\gamma$. Then the distance of the point

(α,β,γ) from the plane $2x-2y+z=\lambda$ is :

(A) 11 (B) 7 (C) 13 (D) 9

Q341. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

Let for $A=\begin{bmatrix}1 & 2 & 3\\ \alpha & 3 & 1\\ 1 & 1 & 2\end{bmatrix}$, $|A|=2$. If $|2\,\mathrm{adj}\,(2\,\mathrm{adj}\,(2\mathrm{A}))|=32^{\mathrm{n}}$, then $3n+\alpha$ is equal to

(A) 11 (B) 9 (C) 12 (D) 10

Q342. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

If $\mathrm{A}=\frac{1}{5!6!7!}\begin{bmatrix}5! & 6! & 7!\\ 6! & 7! & 8!\\ 7! & 8! & 9!\end{bmatrix}$, then $|\mathrm{adj}(\mathrm{adj}(2\,\mathrm{A}))|$ is equal to :

(A) 2^{12} (B) 2^{20} (C) 2^{8} (D) 2^{16}

Q343. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

If A is a 3 × 3 matrix and $|A|=2$, then $|3\,adj\,(|3A|A^2)|$ is equal to :

(A) $3^{12}\,.\,6^{10}$ (B) $3^{11}\,.\,6^{10}$ (C) $3^{12}\,.\,6^{11}$ (D) $3^{10}\,.\,6^{11}$

Q344. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

If a point $\mathrm{P}(\alpha, \beta, \gamma)$ satisfying

$$(\alpha \quad \beta \quad \gamma)\begin{pmatrix}2 & 10 & 8\\ 9 & 3 & 8\\ 8 & 4 & 8\end{pmatrix} = (0 \quad 0 \quad 0)$$

lies on the plane $2x + 4y + 3z = 5$, then $6\alpha + 9\beta + 7\gamma$ is equal to :

(A) $\frac{11}{5}$ (B) 11 (C) -1 (D) $\frac{5}{4}$

Q345. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

If P is a 3×3 real matrix such that $P^T = aP + (a-1)I$, where $a > 1$, then :

(A) $|AdjP| = 1$ (B) $|AdjP| > 1$

(C) $|AdjP| = \frac{1}{2}$ (D) P is a singular matrix

Q346. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

Let $A = \begin{pmatrix}\mathrm{m} & \mathrm{n}\\ \mathrm{p} & \mathrm{q}\end{pmatrix}, \mathrm{d} = |\mathrm{A}| \neq 0$ and $|\mathrm{A} - \mathrm{d}(\mathrm{AdjA})| = 0$. Then

(A) $1 + \mathrm{d}^2 = \mathrm{m}^2 + \mathrm{q}^2$ (B) $1 + d^2 = (m+q)^2$

(C) $(1+d)^2 = m^2 + q^2$ (D) $(1+d)^2 = (m+q)^2$

Q347. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

The set of all values of $\mathrm{t} \in \mathbb{R}$, for which the matrix

$$\begin{bmatrix}e^t & e^{-t}(\sin t - 2\cos t) & e^{-t}(-2\sin t - \cos t)\\ e^t & e^{-t}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t)\\ e^t & e^{-t}\cos t & e^{-t}\sin t\end{bmatrix}$$ is invertible, is :

(A) $\{k\pi, k \in \mathbb{Z}\}$ (B) $\mathbb{R}$

(C) $\left\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\right\}$ (D) $\left\{k\pi + \frac{\pi}{4}, k \in \mathbb{Z}\right\}$

Q348. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

Let $x, y, z > 1$ and $A = \begin{bmatrix}1 & \log_x y & \log_x z\\ \log_y x & 2 & \log_y z\\ \log_z x & \log_z y & 3\end{bmatrix}$. Then $|\mathrm{adj}\ (\mathrm{adj}\ \mathrm{A}^2)|$ is equal to

(A) 6^4 (B) 2^8 (C) 4^8 (D) 2^4

Q349. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

Let A be a 3×3 matrix such that $|\mathrm{adj}(\mathrm{adj}(\mathrm{adj}\ \mathrm{A}))| = 12^4$. Then $|\mathrm{A}^{-1}\ \mathrm{adj}\ \mathrm{A}|$ is equal to

(A) 12 (B) $2\sqrt{3}$ (C) 1 (D) $\sqrt{6}$

Q350. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Let α be a root of the equation $(a-c)x^2+(b-a)x+(c-b)=0$ where a, b, c are distinct real numbers such that the matrix $\begin{bmatrix}\alpha^2 & \alpha & 1\\ 1 & 1 & 1\\ a & b & c\end{bmatrix}$ is singular. Then, the value of $\frac{(a-c)^2}{(b-a)(c-b)}+\frac{(b-a)^2}{(a-c)(c-b)}+\frac{(c-b)^2}{(a-c)(b-a)}$ is

(A) 3 (B) 6 (C) 12 (D) 9

Q351. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

Let the matrix $A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ and the matrix $B_0=A^{49}+2A^{98}$. If $B_n=Adj\,(B_{n-1})$ for all $n\geq 1$, then $\det\,(B_4)$ is equal to :

(A) 3^{28} (B) 3^{30} (C) 3^{32} (D) 3^{36}

Q352. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Let $A=\begin{pmatrix}4 & -2\\ \alpha & \beta\end{pmatrix}$.

If $\mathrm{A}^2+\gamma\mathrm{A}+18\mathrm{I}=\mathrm{O}$, then $\det(\mathrm{A})$ is equal to _____.

(A) -18 (B) 18 (C) -50 (D) 50

Q353. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

Let A be a 2 $\times$ 2 matrix with det (A) = $-$ 1 and det ((A + I) (Adj (A) + I)) = 4. Then the sum of the diagonal elements of A can be :

(A) -1 (B) 2 (C) 1 (D) $-\sqrt{2}$

Q354. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Let A and B be two square matrices of order 2. If $det\,(A)=2$, $det\,(B)=3$ and $\det\left((\det\,(5(\det\,A)B))A^2\right)=2^a3^b5^c$ for some a, b, c, $\in$ N, then a + b + c is equal to :

(A) 10 (B) 12 (C) 13 (D) 14

Q355. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let $A=\begin{pmatrix}2 & -1\\ 0 & 2\end{pmatrix}$. If $B=I-{}^5C_1(adj\,A)+{}^5C_2(adj\,A)^2-\ldots..\ -{}^5C_5(adj\,A)^5$, then the sum of all elements of the matrix B is

(A) -5 (B) -6 (C) -7 (D) -8

Q356. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let A be a matrix of order 3 × 3 and det (A) = 2. Then det (det (A) adj (5 adj (A^3))) is equal to _____.

(A) 512×10^6 (B) 256×10^6 (C) 1024×10^6 (D) 256×10^{11}

Q357. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in R$. Then the sum of the squares of all the values of a, for which $2f'(10) - f'(5) + 100 = 0$, is

(A) 117 (B) 106 (C) 125 (D) 136

Q358. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let for some real numbers α and β, $a = \alpha - i\beta$. If the system of equations $4ix + (1+i)y = 0$ and $8\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)x + \overline{a}y = 0$ has more than one solution, then $\frac{\alpha}{\beta}$ is equal to

(A) $-2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $-2-\sqrt{3}$

Q359. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let A and B be two 3 × 3 matrices such that $AB = I$ and $|A| = \frac{1}{8}$. Then $|adj\,(B\,adj(2A))|$ is equal to

(A) 16 (B) 32 (C) 64 (D) 128

Q360. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let A be a 3 × 3 invertible matrix. If |adj (24A)| = |adj (3 adj (2A))|, then $|A|^2$ is equal to :

(A) 6^6 (B) 2^{12} (C) 2^6 (D) 1

Q361. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Let S = $\{\sqrt{n} : 1 \le n \le 50$ and n is odd$\}$.

Let $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$.

If $\sum\limits_{a \in S} \det(adj\,A) = 100\lambda$, then λ is equal to :

(A) 218 (B) 221 (C) 663 (D) 1717

Q362. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Let A and B be two 3×3 real matrices such that $(A^2 - B^2)$ is invertible matrix. If $A^5 = B^5$ and $A^3B^2 = A^2B^3$, then the value of the determinant of the matrix $A^3 + B^3$ is equal to :

(A) 2 (B) 4 (C) 1 (D) 0

Q363. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

If $A = \begin{pmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{pmatrix}$ and $\det\left(A^2 - \frac{1}{2}I\right) = 0$, then a possible value of α is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Q364. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

Let A be a 3×3 matrix such that

adj A = $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and B = adj(adj A).

If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair,
$(|\lambda|, \mu)$ is equal to :

(A) (3, 81) (B) $\left(9, \frac{1}{9}\right)$
(C) $\left(3, \frac{1}{81}\right)$ (D) $\left(9, \frac{1}{81}\right)$

Q365. JEE Main 2020 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

If the matrices A = $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$,

B = adjA and C = 3A, then $\frac{|adjB|}{|C|}$ is equal to :

(A) 8 (B) 2 (C) 72 (D) 16

Q366. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where i, j = 1, 2, 3. If the determinant of B is 81, then the determinant of A is:

(A) 3 (B) $\frac{1}{3}$ (C) $\frac{1}{9}$ (D) $\frac{1}{81}$

Q367. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

If [x] denotes the greatest integer $\leq$ x, then the system of linear equations [sin θ]x + [−cosθ]y = 0, [cotθ]x + y = 0

(A) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and have infinitely many solutions if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

(B) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and has a unique solution if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

(C) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(D) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

Q368. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Let A and B be two invertible matrices of order 3 $\times$ 3. If det(ABA^T) = 8 and det(AB^{-1}) = 8, then det (BA^{-1} B^T) is equal to :

(A) $\frac{1}{4}$ (B) 16 (C) $\frac{1}{16}$ (D) 1

Q369. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If x = a, y = b, z = c is a solution of the system of linear equations

x + 8y + 7z = 0

9x + 2y + 3z = 0

x + y + z = 0

such that the point (a, b, c) lies on the plane x + 2y + z = 6, then 2a + b + c equals :

(A) -1 (B) 0 (C) 1 (D) 2

Q370. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to :

(A) 4 (B) 11 (C) 5 (D) 0

Q371. AIEEE 2012 · MCQ (Single Correct Answer)

Let P and Q be 3×3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$ then determinant of $\left(P^2 + Q^2\right)$ is equal to :

(A) -2 (B) 1 (C) 0 (D) -1

Q372. AIEEE 2012 · MCQ (Single Correct Answer)

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then $u_1 + u_2$ is equal to :

(A) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$

(B) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$

(C) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

(D) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$

Q373. AIEEE 2009 · MCQ (Single Correct Answer)

Let A be a 2×2 matrix

Statement - 1 : $adj\,(adj\,A) = A$

Statement - 2 : $|adj\,A| = |A|$

(A) statement - 1 is true, statement - 2 is true; statement - 2 is not a correct explanation for statement - 1.

(B) statement - 1 is true, statement - 2 is false.

(C) statement - 1 is false, statement -2 is true

(D) statement -1 is true, statement - 2 is true; statement - 2 is a correct explanation for statement - 1.

Q374. AIEEE 2008 · MCQ (Single Correct Answer)

Let A be a square matrix all of whose entries are integers.
Then which one of the following is true?

(A) If det $A = \pm 1$, then A^{-1} exists but all its entries are not necessarily integers

(B) If det $A \neq \pm 1$, then A^{-1} exists and all its entries are non integers

(C) If det $A = \pm 1$, then A^{-1} exists but all its entries are integers

(D) If det $A = \pm 1$, then A^{-1} need not exists

Q375. AIEEE 2007 · MCQ (Single Correct Answer)

Let $A = \begin{vmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{vmatrix}$. If $|A^2| = 25$, then $|\alpha|$ equals

(A) $1/5$

(B) 5

(C) 5^2

(D) 1

Q376. JEE Main 2023 (Online) 31st January Evening Shift · Numerical

Let A be a $n \times n$ matrix such that $|\mathrm{A}| = 2$. If the determinant of the matrix $\operatorname{Adj}\left(2 \cdot \operatorname{Adj}\left(2\,\mathrm{A}^{-1}\right)\right)$ is 2^{84}, then n is equal to :

Q377. JEE Main 2022 (Online) 27th July Evening Shift · Numerical

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers.

If $\frac{\det(\operatorname{adj}(\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A))))}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3 - tuples (α, β, γ) is _____.

Q378. JEE Main 2022 (Online) 27th June Morning Shift · Numerical

The positive value of the determinant of the matrix A, whose

Adj(Adj(A)) = $\begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix}$, is _____.

Q379. JEE Main 2021 (Online) 26th August Evening Shift · Numerical

Let A be a 3 × 3 real matrix. If det(2Adj(2 Adj(Adj(2A)))) = 2^{41}, then the value of det(A^2) is equal to _____.

Q380. JEE Main 2021 (Online) 25th July Morning Shift · Numerical

Let $M = \left\{A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \{\pm 3, \pm 2, \pm 1, 0\}\right\}$. Define f : M → Z, as f(A) = det(A), for all A∈M, where z is set of all integers. Then the number of A∈M such that f(A) = 15 is equal to _____.

Q381. JEE Main 2021 (Online) 20th July Evening Shift · Numerical

Let $A = \{a_{ij}\}$ be a 3 × 3 matrix,

where $a_{ij} = \begin{cases} (-1)^{j-i} & if \quad i < j, \\ 2 & if \quad i = j, \\ (-1)^{i+j} & if \quad i > j \end{cases}$

then $\det(3Adj(2A^{-1}))$ is equal to _____.

Q382. JEE Main 2021 (Online) 17th March Morning Shift · Numerical

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of det($A^4$) + det($A^{10} - (\text{Adj}(2A))^{10}$) is equal to _____.

Q383. JEE Main 2021 (Online) 16th March Morning Shift · **Numerical**

Let $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and

$A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega+1 \end{bmatrix}$ where

$\omega = \frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $(P^{-1}AP-I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____.

Q384. JEE Main 2021 (Online) 25th February Morning Shift · **Numerical**

Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x, y and z are real numbers such that x + y + z > 0 and xyz = 2. If $A^2 = I_3$, then the value of $x^3 + y^3 + z^3$ is _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **C**	3. **D**	4. **C**	5. **A**
6. **B**	7. **D**	8. **B**	9. **B**	10. **C**
11. **D**	12. **D**	13. **B**	14. **B**	15. **B**
16. **C**	17. **B**	18. **B**	19. **B**	20. **B**
21. **A**	22. **B**	23. **D**	24. **C**	25. **C**
26. **A**	27. **C**	28. **C**	29. **D**	30. **A**
31. **D**	32. **B**	33. **B**	34. **B**	35. **B**
36. **D**	37. **A**	38. **D**	39. **D**	40. **C**
41. **D**	42. **A**	43. **D**	44. **B**	45. **C**
46. **C**	47. **B**	48. **B**	49. **D**	50. **D**
51. **D**	52. **D**	53. **B**	54. **B**	55. **D**
56. **D**	57. **C**	58. **B**	59. **A**	60. **C**
61. **D**	62. **C**	63. **C**	64. **D**	65. **A**
66. **A**	67. **B**	68. **B**	69. **B**	70. **C**
71. **D**	72. **D**	73. **C**	74. **D**	75. **D**
76. **D**	77. **C**	78. **C**	79. **C**	80. **A**
81. **B**	82. **B**	83. **C**	84. **C**	85. **B**
86. **B**	87. **D**	88. **A**	89. **D**	90. **D**
91. **C**	92. **D**	93. **C**	94. **A**	95. **D**
96. **A**	97. **B**	98. **B**	99. **D**	100. **B**
101. **B**	102. **D**	103. **D**	104. **D**	105. **D**
106. **C**	107. **A**	108. **C**	109. **C**	110. **D**
111. **C**	112. **C**	113. **C**	114. **B**	115. **C**
116. **D**	117. **D**	118. **D**	119. **C**	120. **B**
121. **A**	122. **B**	123. **A**	124. **B**	125. **D**
126. **C**	127. **C**	128. **D**	129. **B**	130. **A**
131. **C**	132. **B**	133. **C**	134. **B**	135. **D**
136. **C**	137. **B**	138. **A**	139. **A**	140. **A**
141. **B**	142. **B**	143. **D**	144. **B**	145. **A**
146. **A**	147. **D**	148. **C**	149. **B**	150. **A**

151. **A**	152. **D**	153. **B**	154. **B**	155. **A**
156. **C**	157. **A**	158. **D**	159. **C**	160. **B**
161. **A**	162. **A**	163. **C**	164. **C**	165. **C**
166. **D**	167. **B**	168. **A**	169. **B**	170. **A**
171. **B**	172. **D**	173. **B**	174. **B**	175. **C**
176. **C**	177. **D**	178. **B**	179. **B**	180. **C**
181. **B**	182. **D**	183. **C**	184. **A**	185. **C**
186. **B**	187. **C**	188. **B**	189. **D**	190. **A**
191. **B**	192. **D**	193. **D**	194. **B**	195. **B**
196. **B**	197. **A**	198. **C**	199. **B**	200. **C**
201. **C**	202. **A**	203. **B**	204. **D**	205. **C**
206. **A**	207. **C**	208. **D**	209. **D**	210. **A**
211. **A**	212. **A**	213. **D**	214. **A**	215. **A**
216. **C**	217. **C**	218. **A**	219. **C**	220. **C**
221. **C**	222. **A**	223. **C**	224. **D**	225. **A**
226. **B**	227. **D**	228. **D**	229. **B**	230. **B**
231. **D**	232. **C**	233. **A**	234. **D**	235. **C**
236. **D**	237. **D**	238. **C**	239. **D**	240. **D**
241. **B**	242. **A**	243. **A**	244. **D**	245. **B**
246. **A**	247. **A**	248. **B**	249. **C**	250. **C**
251. **B**	252. **D**	253. **D**	254. **D**	255. **B**
256. **D**	257. **D**	258. **A**	259. **D**	260. **B**
261. **A**	262. **A**	263. **D**	264. **D**	265. **B**
266. **C**	267. **C**			

Numerical

268. **4**	269. **34**	270. **0**	271. **312**	272. **3**
273. **62**	274. **18**	275. **225**	276. **36**	277. **6**
278. **38 or -38**	279. **2**	280. **1613**	281. **44**	282. **34**
283. **450**	284. **14**	285. **7**	286. **38**	287. **55**
288. **5**	289. **25**	290. **27**	291. **2**	292. **7**
293. **113**	294. **10**	295. **28**	296. **6**	297. **2**
298. **24**	299. **5**	300. **495**	301. **10**	302. **4**
303. **2**	304. **5376**	305. **50**	306. **24**	307. **17**

308. **1**	309. **58**	310. **25**	311. **100**	312. **1**
313. **100**	314. **8**	315. **5**	316. **2020**	317. **5**
318. **6**	319. **11**	320. **3125**	321. **910**	322. **1**
323. **6**	324. **2020**	325. **2**	326. **1**	327. **766**
328. **4**	329. **13**	330. **21**	331. **17**	332. **540**
333. **3**	334. **5**	335. **8**	336. **10**	337. **672**
338. **13**				

Out of syllabus

339. **A**	340. **B**	341. **A**	342. **D**	343. **B**
344. **B**	345. **A**	346. **D**	347. **B**	348. **B**
349. **B**	350. **A**	351. **C**	352. **B**	353. **B**
354. **B**	355. **C**	356. **A**	357. **C**	358. **B**
359. **C**	360. **C**	361. **B**	362. **D**	363. **A**
364. **C**	365. **A**	366. **C**	367. **B**	368. **C**
369. **C**	370. **B**	371. **C**	372. **D**	373. **D**
374. **C**	375. **A**	376. **5**	377. **42**	378. **14**
379. **4**	380. **16**	381. **108**	382. **16**	383. **36**
384. **7**				